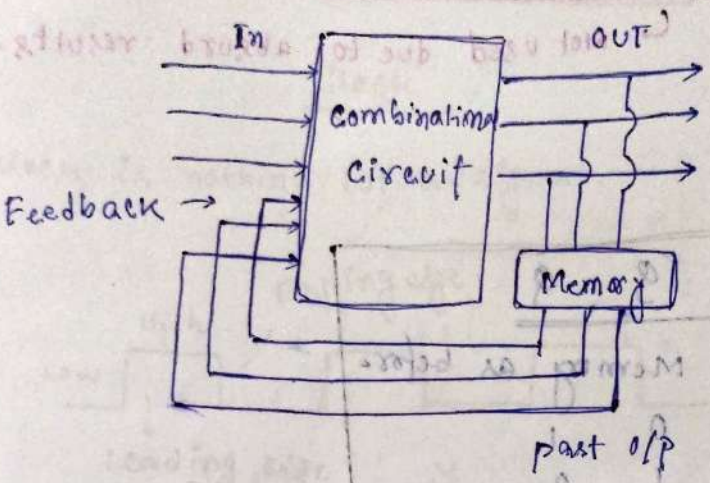
# Sequential Circuit

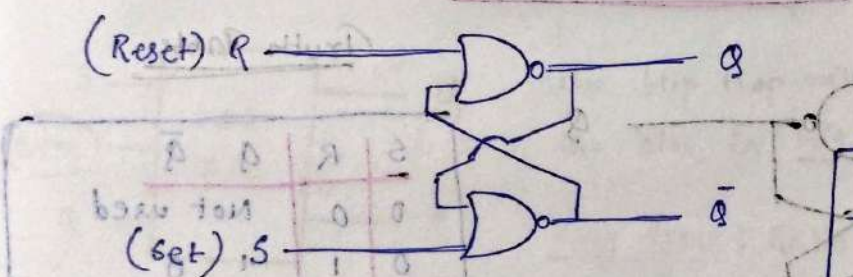
In sequential circuits, the present output depends on the present input as well as past input/outputs.

Flip flop, counters, Registers.




Flip Flops
(keeps the output until we need to use it).

The basic storage element is called LATCH. As the name suggests it latches "0" or "1".




Reset, Q=0
Set, Q=1

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

NOR gate truth table.
whenever A=1, Y=0

**Case 1:** S=0, R=1, Q=0, $\bar{Q}$=1

S=0, R=0, Q=0, $\bar{Q}$=1

Memory.

**case 2:** S=1, R=0, Q=1, $\bar{Q}$=0

(Here S=1, so $\bar{Q}$=0 and R(0,0)→1, Q=1)

S=0, R=0, Q=1, $\bar{Q}$=0

↳ Memory.

**Case III :** $S=1,\ R=1,\ Q=0,\ \bar{Q}=0$

$Q$ cannot be equal to $\bar{Q}$

$S=0,\ R=0,\ Q=0,\ \bar{Q}=1$

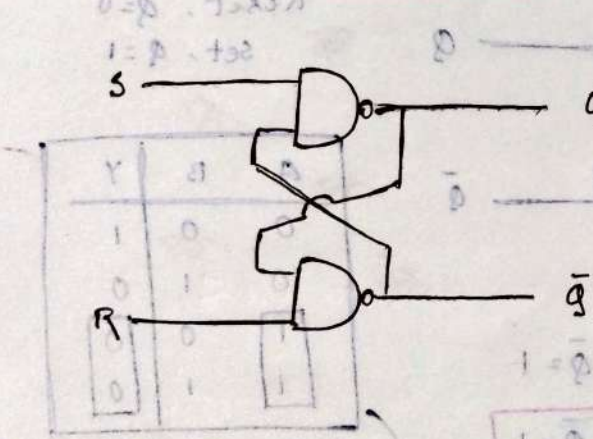
$Q=1,\ \bar{Q}=0$

↳ Not used due to absurd results.

Truth table →

| S | R | Q | $\bar{Q}$ |
|---|---|---|---|
| 0 | 0 | Memory as before. | |
| 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | Not used | |

## NAND S-R LATCH




Truth Table

| S | R | Q | $\bar{Q}$ |
|---|---|---|---|
| 0 | 0 | Not used | |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | Memory store | |

**Case 1 :** when $S=0,\ R=0,\ Q=1,\ \bar{Q}=1$

But $Q \neq \bar{Q}$. This situation is clearly absurd and not possible.

and, $S=1,\ R=1$

if we view $\bar{Q}$ first; $Q=0,\ \bar{Q}=1$

and if we view $Q$ first, $Q=1,\ \bar{Q}=0$

This is a discrepancy. so, it is not possible.

NAND Truth Table

| A | B | Y |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**Case 2:** $S=0,\ R=1,\ Q=0,\ \bar{Q}=1$

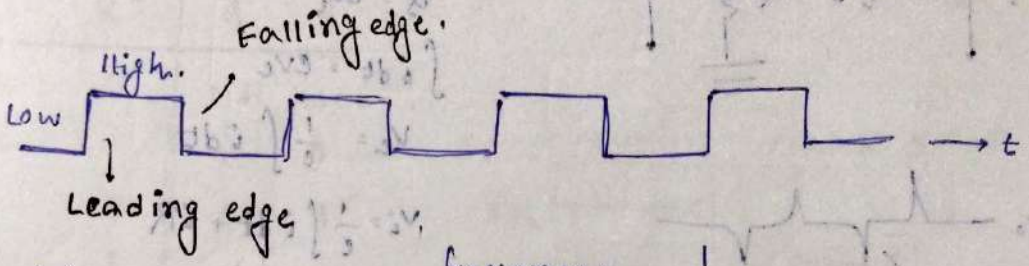
$S=0,\ R=0,\ Q=1,\ \bar{Q}=0$ Memory.

**Case 3:** $S=1,\ R=0,\ Q=0,\ \bar{Q}=1$

$S=0,\ R=0,\ Q=0,\ \bar{Q}=1$ Memory.

## Clock

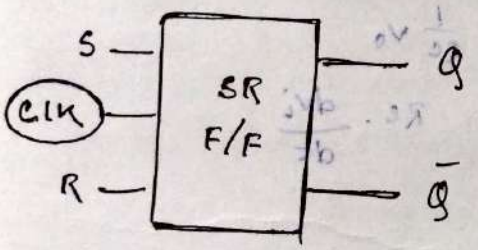
A clock is nothing but a signal.




50 % duty cycle.

frequency = $\frac{1}{\text{Time period}}$

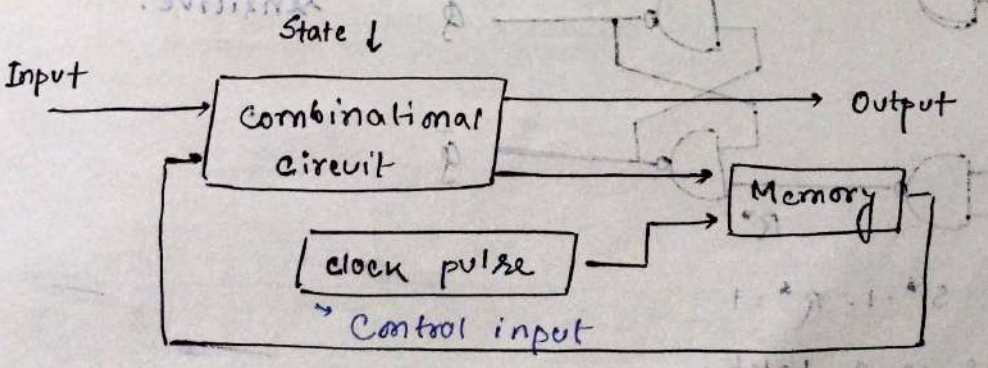
$$f = \frac{1}{T}$$




The flip flop will work only when the clk is high.

Duty cycle: Ratio of time for which the signal is high to total time.

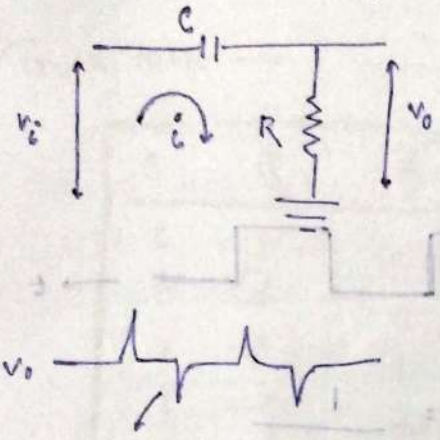
## Triggering Methods.



2 type of triggering — Level triggering, Edge triggering

Edge triggering → Positive edge triggering, Negative edge triggering.

Differentiator ckt.




$v_i - v_c - v_o = 0$

$v_i = v_c + v_o$

$q = cv_c$

$\int i\,dt = cv_c$

$v_c = \frac{1}{c}\int i\,dt$

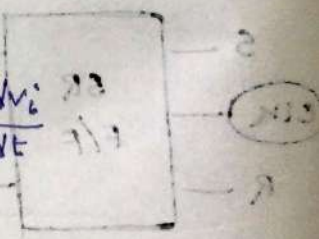
$v_i = \frac{1}{c}\int i\,dt + iR$

$v_i = \frac{1}{c}\int i\,dt$

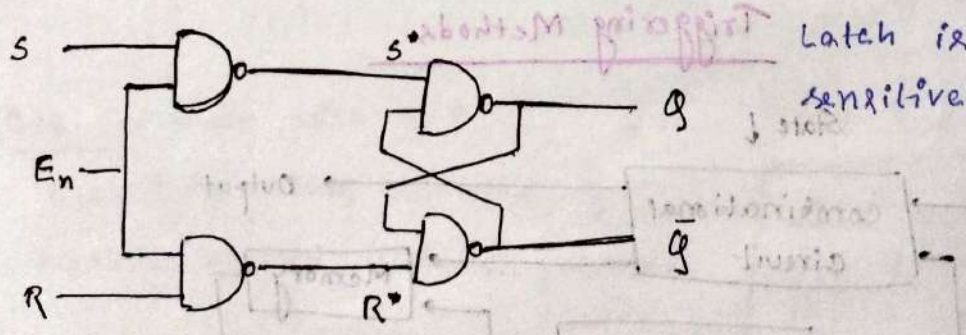
$v_i = \frac{1}{c}\int \frac{v_o}{R}\,dt$

$\frac{dv_i}{dt} = \frac{1}{RC} v_o$

$v_o = RC \cdot \frac{dv_i}{dt}$



## Difference between Latch and F/F

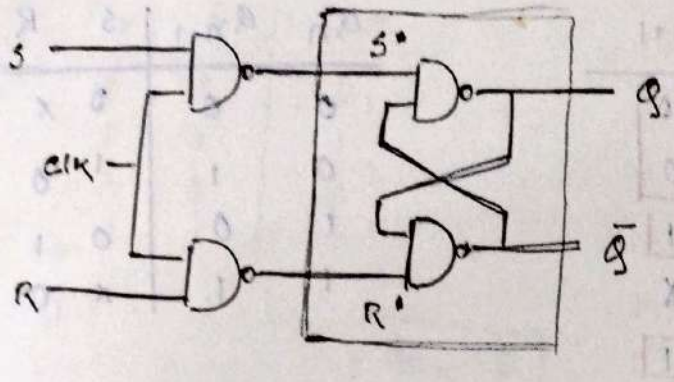



Latch is level sensitive.

$E_n = 1$, $S^* = 1$, $R^* = 1$ acts as a Latch.

For Flip Flop in place of enable clock is present.

# SR Flip Flop

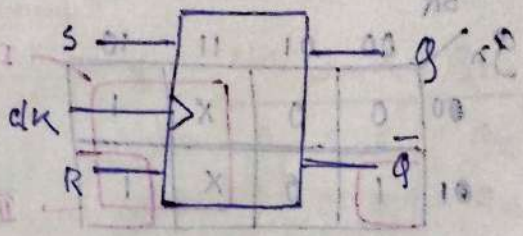



$S^* = \overline{S \cdot CLK} = \bar{S} + \overline{CLK}$

$R^* = \overline{R \cdot CLK} = \bar{R} + \overline{CLK}$

## SR Latch with NAND

| $S^*$ | $R^*$ | $Q$ | $\bar{Q}$ |
|---|---|---|---|
| 0 | 0 | Not used | |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | Memory | |




| CLK | S | R | $Q_{n+1}$ | $\bar{Q}_{n+1}$ |
|---|---|---|---|---|
| 0 | X | X | Memory | |
| 1 | 0 | 0 | Memory | |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | Not used | |

## Characteristic table and excitation table of SR F/F

### Truth table

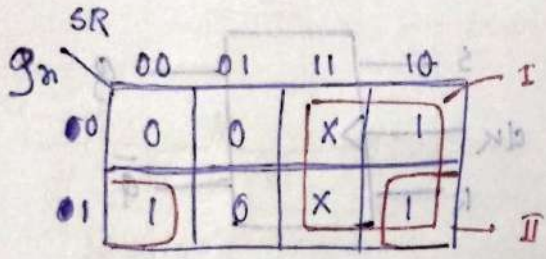
| CLK | S | R | $Q_{n+1}$ |
|---|---|---|---|
| 0 | X | X | $Q_n$ |
| 1 | 0 | 0 | $Q_n$ |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | Invalid. |

## Characteristic Table

| $Q_n$ | S | R | $Q_{n+1}$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | X |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | X |

## Excitation Table

| $Q_n$ | $Q_{n+1}$ | S | R |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | X | 0 |



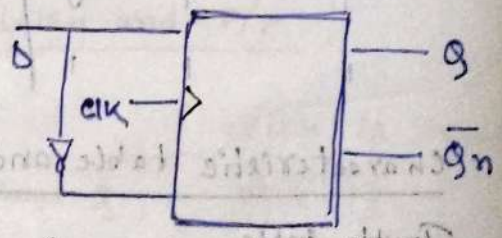

$$Q_{n+1} = S + Q_n \bar{R}$$

## D Flip Flop

### Truth table of SR F/F

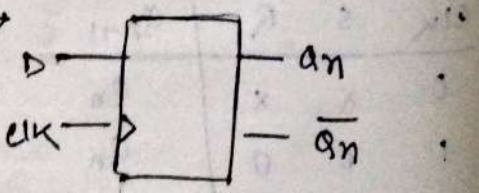
| clk | S | R | $Q_{n+1}$ |
|---|---|---|---|
| 0 | X | X | Memory ($Q_n$) |
| 1 | 0 | 0 | " |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | Invalid. |

S and R are complement to each other




### Truth table of D F/F

| clk | D | $Q_{n+1}$ |
|---|---|---|
| 0 | X | $Q_n$ |
| 1 | 0 | 0 |
| 1 | 1 | 1 |



## Characteristics table

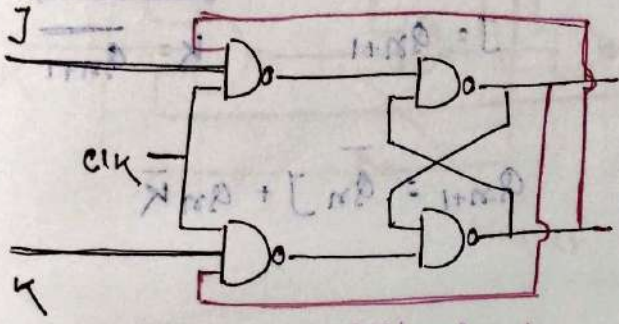
| $Q_n$ | D | $Q_{n+1}$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

$Q_{n+1} = D$

## Excitation Table

| $Q_n$ | $Q_{n+1}$ | D |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

## JK Flip Flop




## Truth table of JK

| clk | J | K | $Q_{n+1}$ |
|---|---|---|---|
| 0 | x | x | $Q_n$ |
| 1 | 0 | 0 | $Q_n$ |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | $\bar{Q}_n$ (Toggle) |

clk = 0 → Memory.

clk = 1, J = 1, K = 0, Q = 1, $\bar{Q}$ = 0

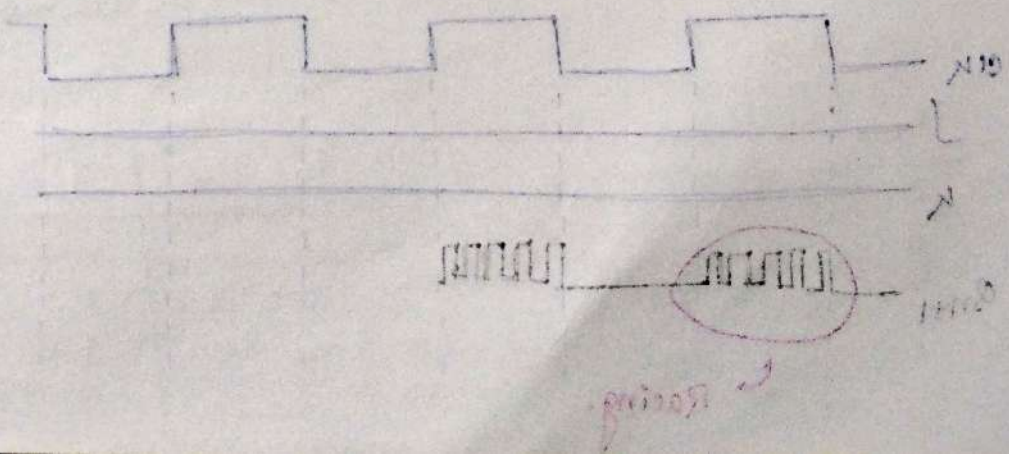
clk = 1, J = 0, K = 0, Q = 0, $\bar{Q}$ = 1

clk = 1, J = 1, K = 1

assume, Q = 0, $\bar{Q}$ = 1

Q = 0, 1, 0, 1, ..

$\bar{Q}$ = 1, 0, 1, 0, ..

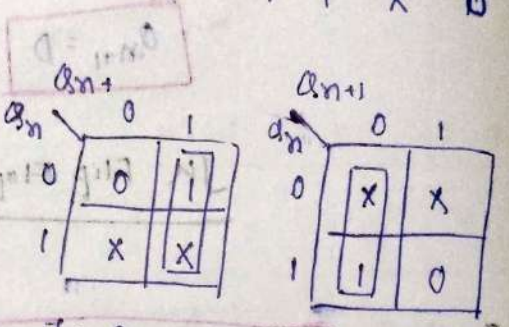


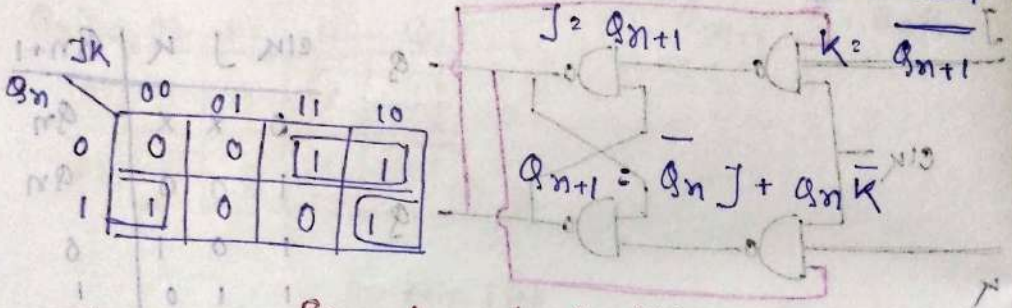
## Characteristics Table

| $Q_n$ | J | K | $Q_{n+1}$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

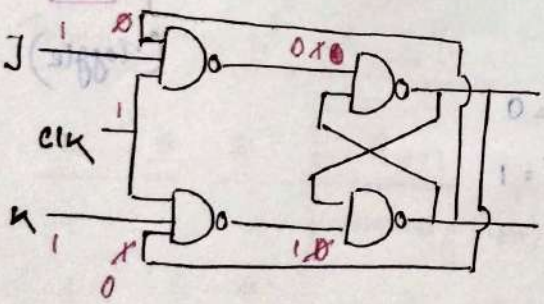
## Excitation Table

| $Q_n$ | $Q_{n+1}$ | J | K |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |




## Race Around Condition

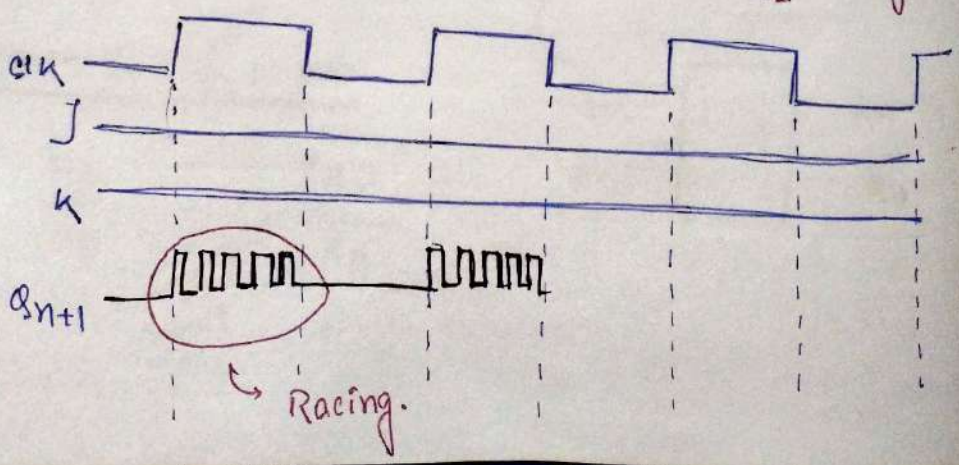



| Clk | J | K | $Q_{n+1}$ |
|---|---|---|---|
| 0 | X | X | $Q_n$ |
| 1 | 0 | 0 | $Q_n$ |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | $\overline{Q_n}$ |

Toggle → controlled

Race around → Not controlled

$T_{(2)}$ > delay.



condition to overcome Racing –

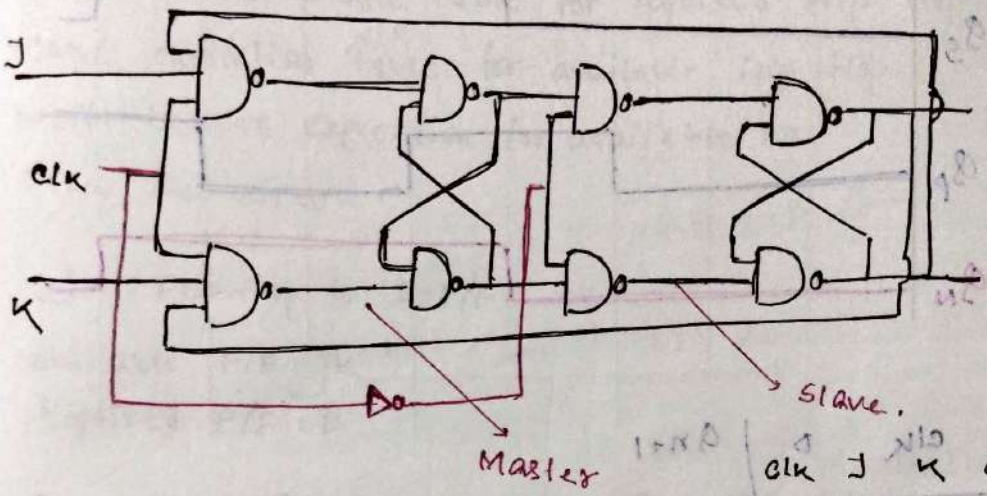
I. $T_{/2}$ < propagation delay of Flip flop

II. Edge triggering.

** III. Master slave.

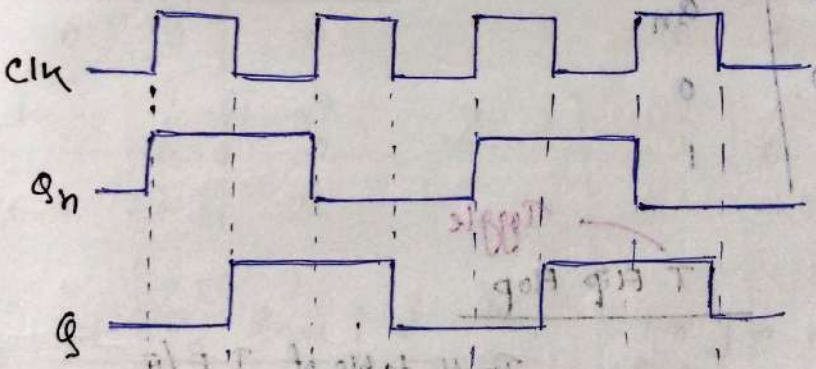
{ Master slave operation is same as -ve edge triggering.

## Master - slave operation of JK F/F

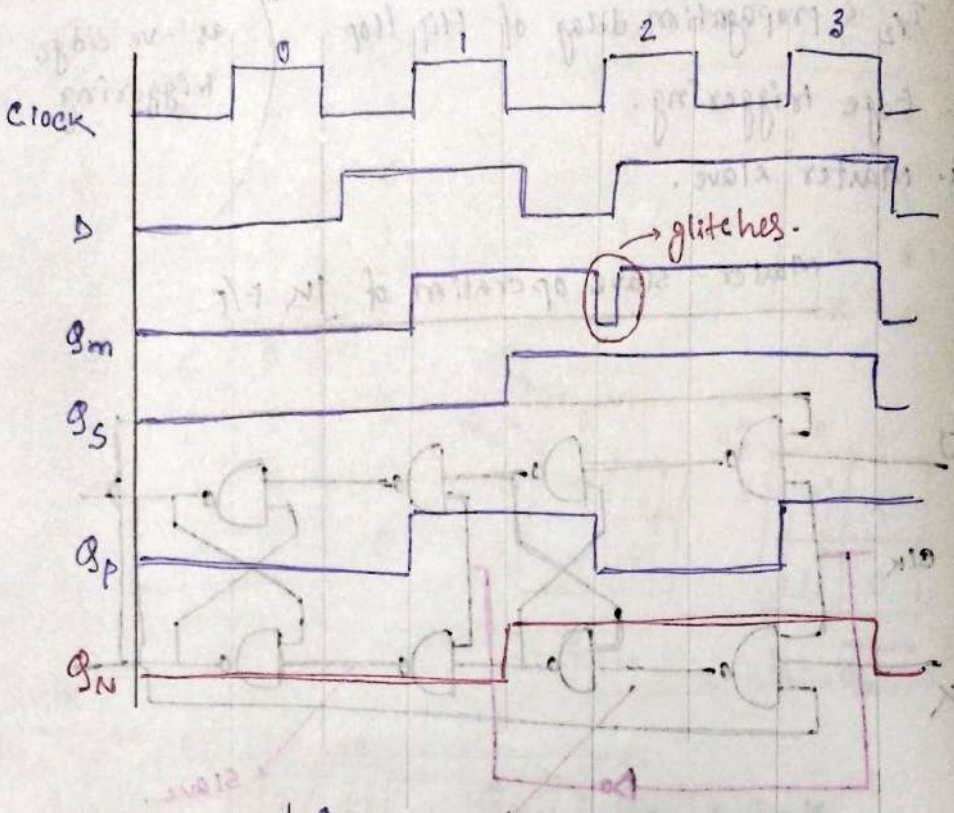



| clk | J | K | $Q_{n-1}$ | |
|---|---|---|---|---|
| 1 | 0 | 0 | | $Q_n$ |
| 1 | 0 | 1 | | 0 |
| 1 | 1 | 0 | | 1 |
| 1 | 1 | 1 | | $\bar{Q}_n$ |

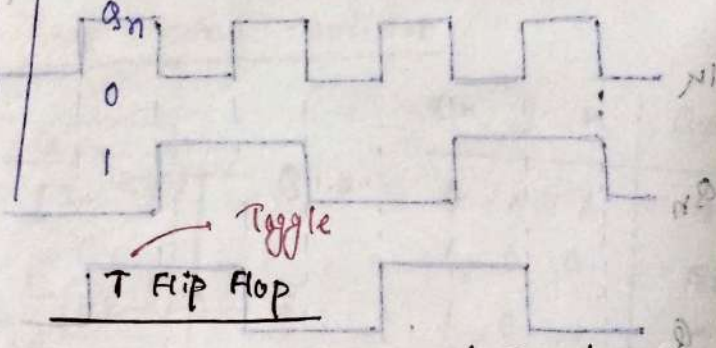
Toggling



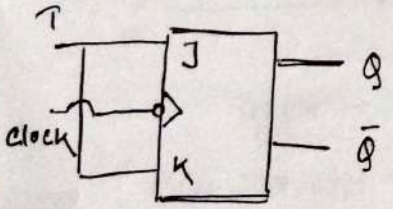
## Behaviour of Master slave D flip flop




| Clk | D | $Q_{n+1}$ |
|---|---|---|
| 0 | x | $Q_n$ |
| 1 | 0 | 0 |
| 1 | 1 | 1 |




## T Flip Flop




### Truth table of T F/F

| Clk | T | $Q_{n+1}$ |
|---|---|---|
| 0 | x | $Q_n$ (Memory) |
| 1 | 0 | $Q_n$ (Memory) |
| 1 | 1 | $\bar{Q}_n$ (Toggling). |

### Ch. Table

| $Q_n$ | T | $Q_{n+1}$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

Odd 1's detector XOR

Excitation Table

| $Q_n$ | $Q_{n+1}$ | T |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$Q_{n+1} = Q_n \oplus T$

## Flip Flop Conversion

Steps :

1. Identify available and required flip flop.
2. Make characteristic table for required flip flop.
3. Make excitation table for available flip flop.
4. write boolean expression for available FF.
5. Draw the circuit.

Ex. JK Flip Flop to D-F/F

available F/F : JK

Required F/F = D

| $Q_n$ | D | $Q_{n+1}$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

| $Q_n$ | $Q_{n+1}$ | J | K |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |

| $Q_n$ | D | $Q_{n+1}$ | J | K |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | X |
| 0 | 1 | 1 | 1 | X |
| 1 | 0 | 0 | X | 1 |
| 1 | 1 | 1 | X | 0 |

For J →

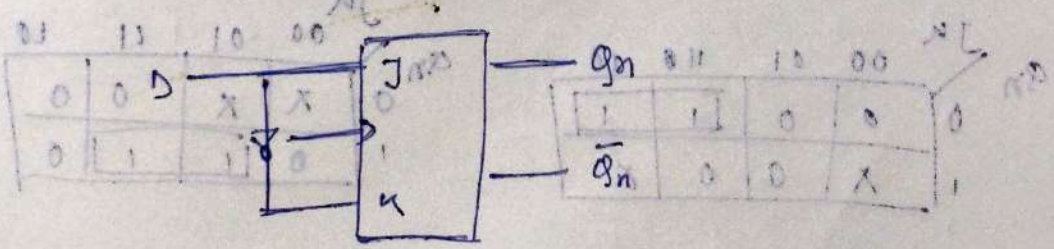
| $Q_n$ \ D | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | X | X |

$J = D$

For K →

| $Q_n$ \ D | 0 | 1 |
|---|---|---|
| 0 | X | X |
| 1 | 1 | 0 |

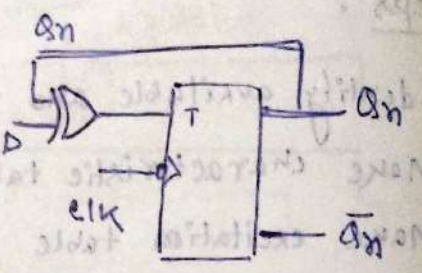
$K = \bar{D}$



# T F/F to D F/F

available flip flop → T
Required flip flop → D

| $Q_n$ | D | $Q_{n+1}$ | T |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |



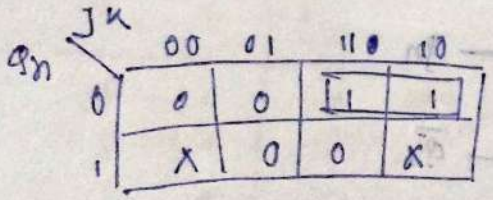

$T = D \oplus Q_n$

# SR F/F to JK F/F

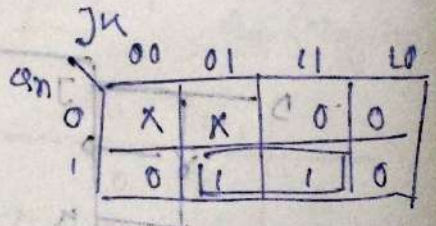
available F/F = SR
Required F/F = JK

| $Q_n$ | J | K | $Q_{n+1}$ | S | R |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | X |
| 0 | 0 | 1 | 0 | 0 | X |
| 0 | 1 | 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | X | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 1 | 0 | 1 | X | 0 |
| 1 | 1 | 1 | 0 | 0 | 1 |

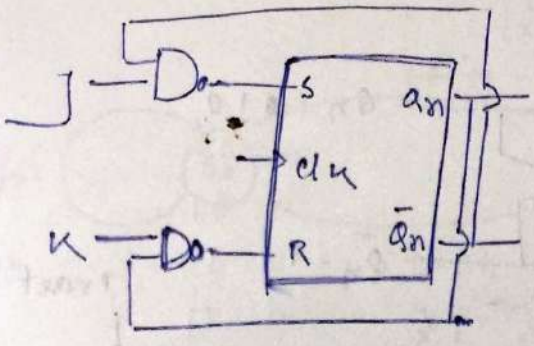
| $Q_n$ | $Q_{n+1}$ | S | R |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | X | 0 |




$S = \bar{Q}_n J$




$R = Q_n K$




## SR F/F to T F/F

Available F/F : SR
Required F/F : T

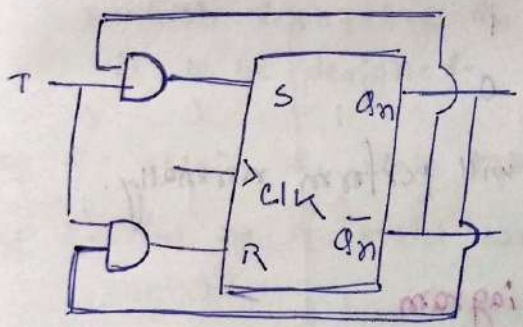
| $Q_n$ | T | $Q_{n+1}$ | S | R |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | X |
| 0 | 1 | 1 | 1 | 0 |
| 1 | 0 | 1 | X | 0 |
| 1 | 1 | 0 | 0 | 1 |

| $Q_n$ \ T | 0 | 1 |
|---|---|---|
| 0 | 0 | (1) |
| 1 | X | 0 |

$S = \bar{Q}_n T$

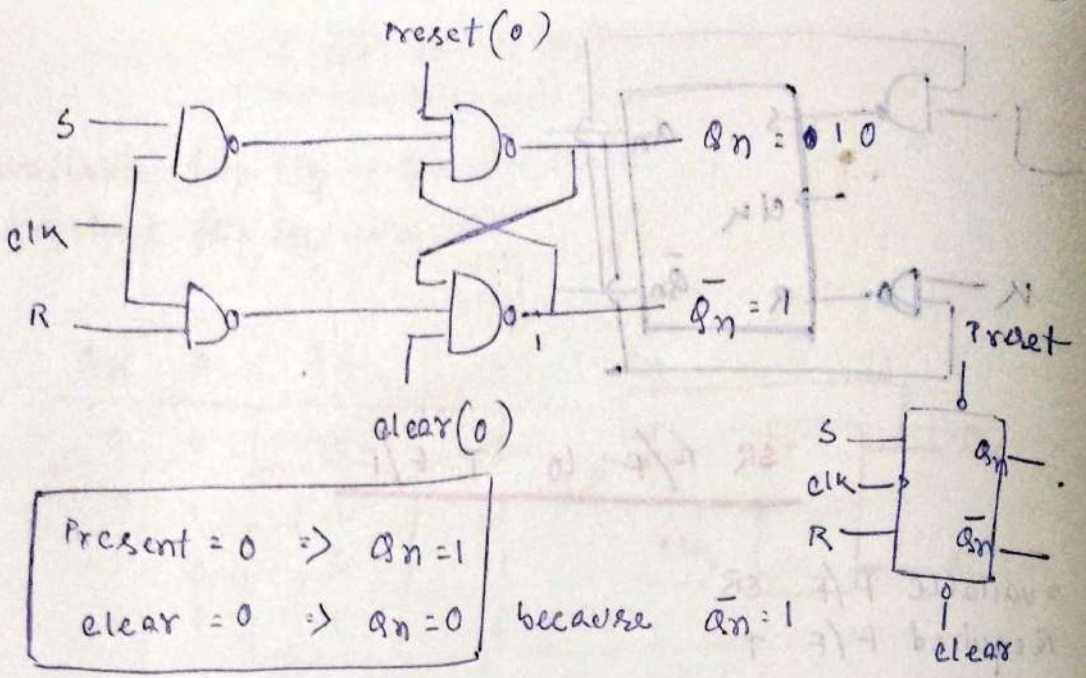
| $Q_n$ \ T | 0 | 1 |
|---|---|---|
| 0 | X | 0 |
| 1 | 0 | (1) |

$R = Q_n T$




## Preset and clear Inputs

- They are the direct inputs or overriding inputs or asynchronous inputs.
- The synchronous inputs are S, R, J, K, D & T.




| | | |
|---|---|---|
| Present = 0 | ⇒ | Qn = 1 |
| clear = 0 | ⇒ | Qn = 0 because $\bar{Q}_n = 1$ |

whatever be the value of clock and synchronous inputs

| Preset | clear | Qn |
|---|---|---|
| 0 | 0 | Not used |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | F/F will perform normally. |

## STATE Diagram

For JK F/F :

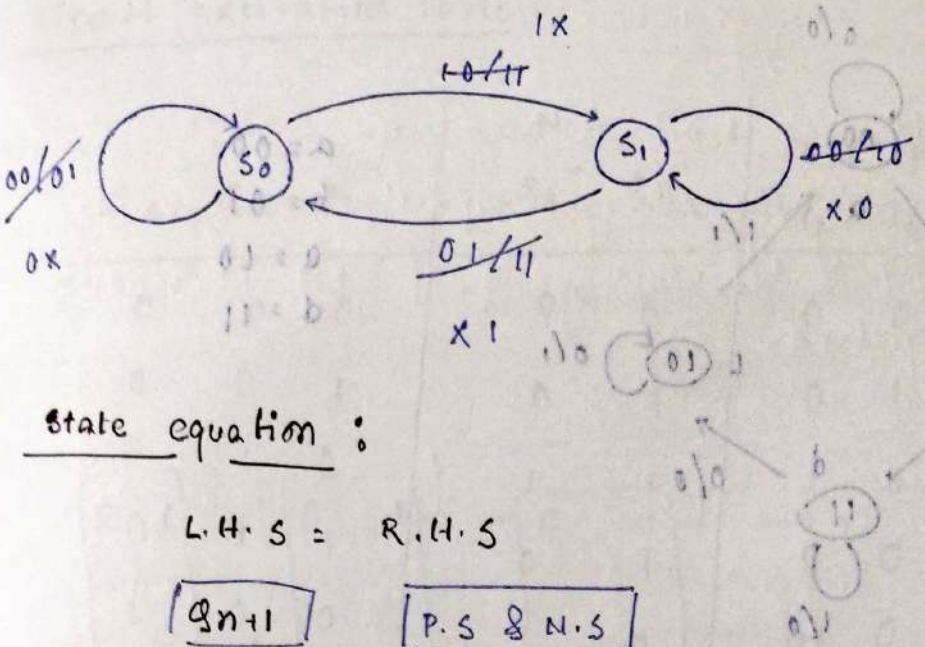
| P.S | i/p | | N.S |
|---|---|---|---|
| Qn | J | K | Qn+1 |
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

1 FF

$S_0$, $2^1 = 2$ states

$S_0 = 0$

$S_1 = 1$




State equation :

L.H.S = R.H.S

$Q_{n+1}$ | P.S & N.S

$$Q_{n+1} = \bar{Q}_n J + Q_n \bar{K}.$$

## Design procedure for clocked sequential cuts

Step 1. A state diagram or timing diagram is given, which describes the behaviour of the circuit that is to be designed.

Step 2. Obtain the state table

Step 3. The no. of states can be reduced by state reduction method.

Step 4. Do state assignment (if required).

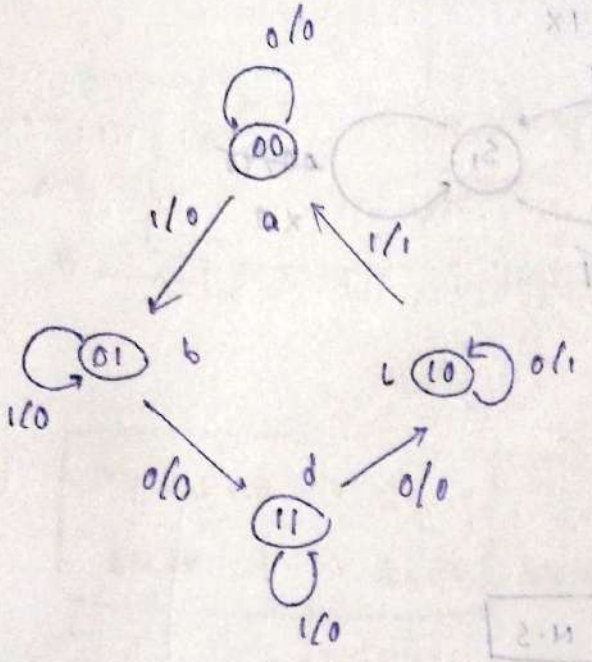
Step 5. Determine the no. of F/F required and assign letter symbols.

Step 6. Decide the type of F/F to be used.

Step 7. Derive the circuit excitation table from state table.

Step 8. Obtain the expression for circuit output and F/F input.

Step 9. Implement the circuit.




State table :

| P.S | | N.S | | | | O/P | |
|---|---|---|---|---|---|---|---|
| | | x=0 | | x=1 | | | |
| $Q_A$ | $Q_B$ | $Q_A^+$ | $Q_B^+$ | $Q_A^+$ | $Q_B^+$ | x=0 | x=1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 |

- No state reduction possible (The next state and o/p should be same to reduce state).

→ 2 F/F requird. (A) (B)

circuit excitation table :

| P.S | | | N.S | | F/F i/p | | |
|---|---|---|---|---|---|---|---|
| $Q_A$ | $Q_B$ | $x$ | $Q_A^+$ | $Q_B^+$ | $T_A$ | $T_B$ | $J$ |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |

| $Q_n$ | $Q_{n+1}$ | $T_n$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

$T_A = \bar{Q}_A Q_B x + Q_A \bar{Q}_B \bar{x}$

$T_B = \bar{Q}_A \bar{Q}_B x + Q_A Q_B \bar{x}$

$J = Q_A \bar{Q}_B \bar{x} + Q_A \bar{Q}_B x$

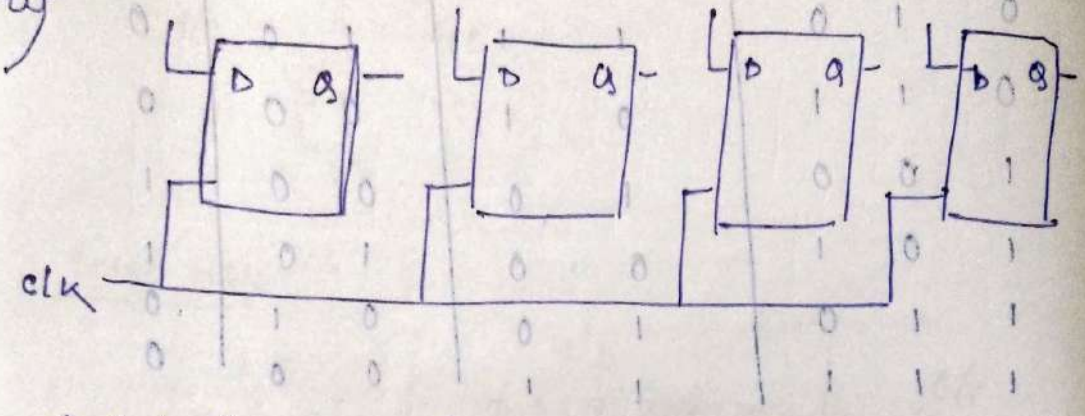
$= Q_A \bar{Q}_B (x + \bar{x})$

$= Q_A \cdot \bar{Q}_B$

# Registers

→ Flip Flop is 1 bit memory cell.

→ To increase the storage capacity, we have to use group of F/F. This group of F/F is known as registers.




# Data in serial form → temporal code

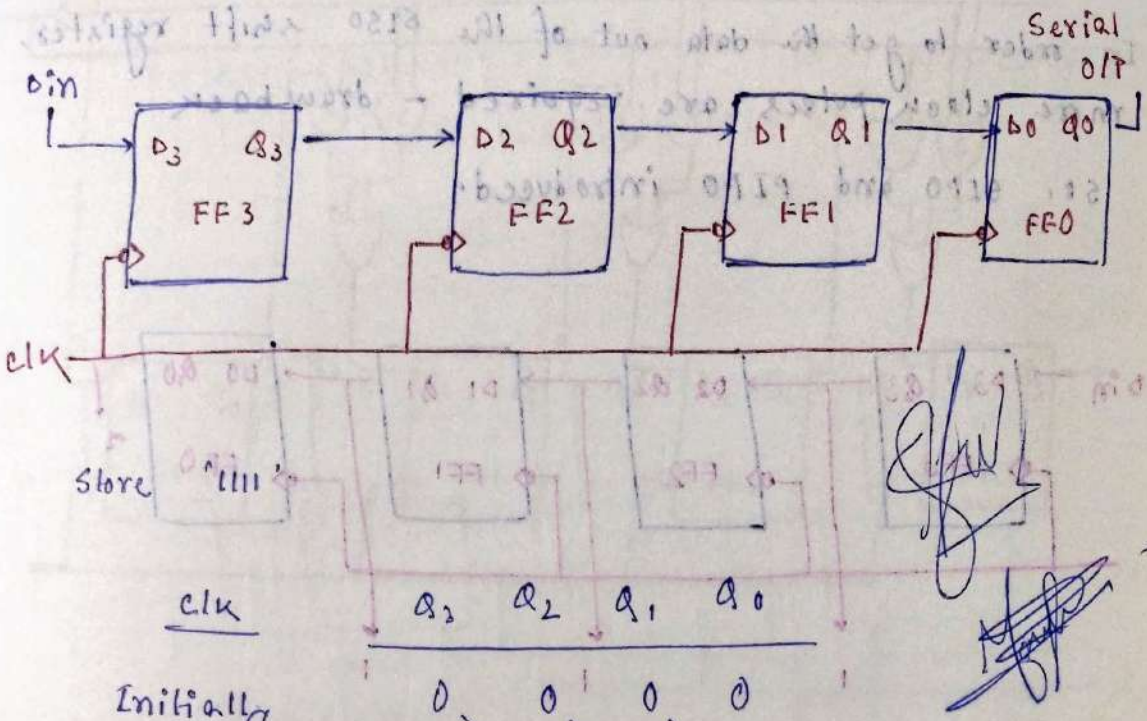
# Data in parallel form → special code.

## Classification of registers:

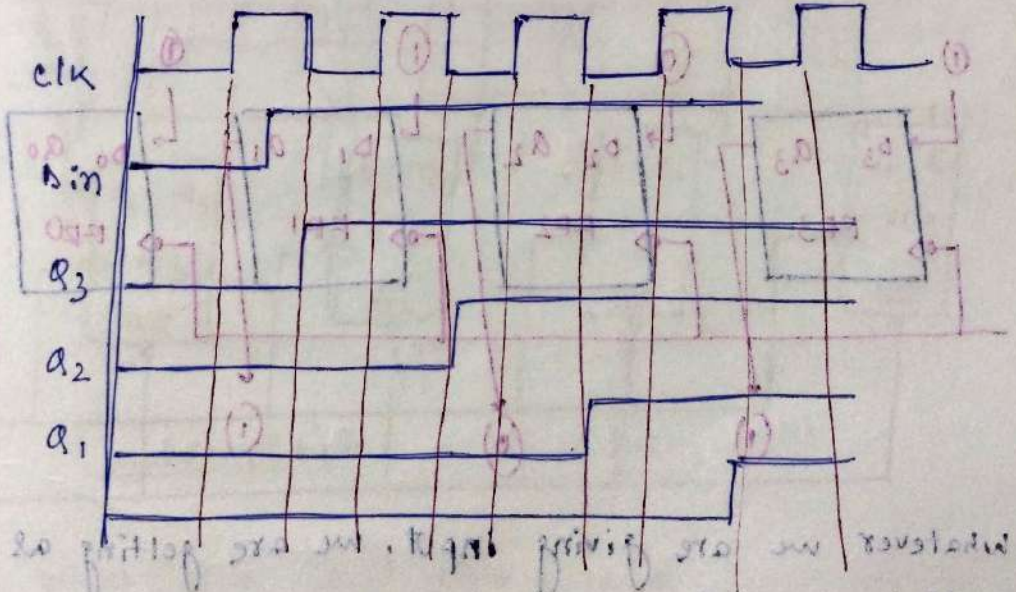
1. Depending on i/p & o/p:
   - (i) SISO (Serial in serial out).
   - (ii) ~~PI SO~~ SIPO (serial in parallel out).
   - (iii) ~~SIPO (serial in~~
   - (iii) PISO (Parallel in serial out).
   - (iv) PIPO (Parallel in parallel out).

2. Depending on application:
   - * (i) Shift register
   - (ii) Storage register.

# shift register (Serial in Serial out)




| clk | $Q_3$ | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|---|
| Initially | 0 | 0 | 0 | 0 |
| ↓ 1 | 1 | 0 | 0 | 0 |
| ↓ 2 | 1 | 1 | 0 | 0 |
| ↓ 3 | 1 | 1 | 1 | 0 |
| ↓ 4. | | | | |

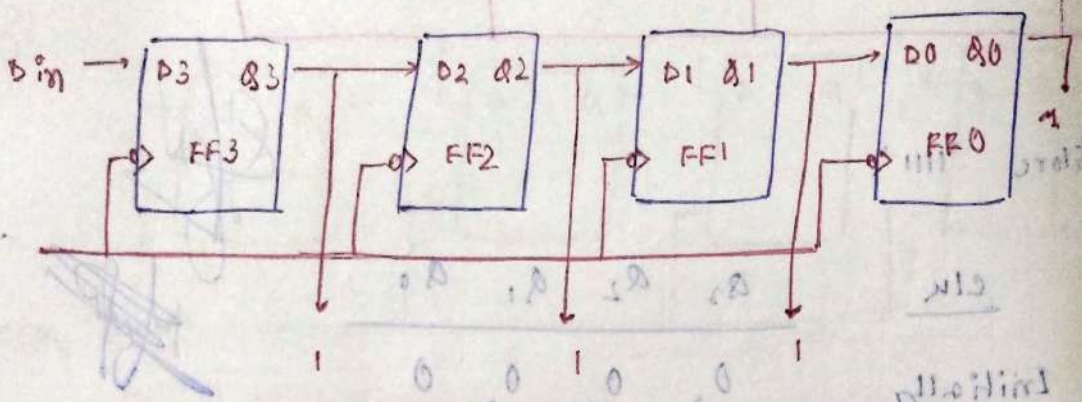



used in long distance transmission.

## SIPO and PIPO mode

In order to get the data out of the SISO shift register, more clock pulses are required - drawback

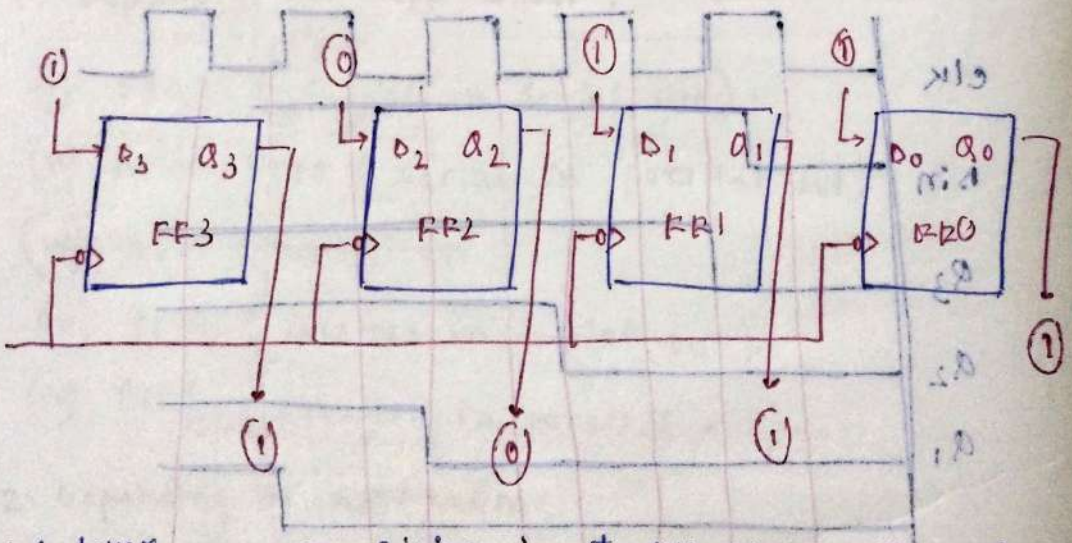
So, SIPO and PIPO introduced.




Here we are getting the data with in 4 clock pulses.

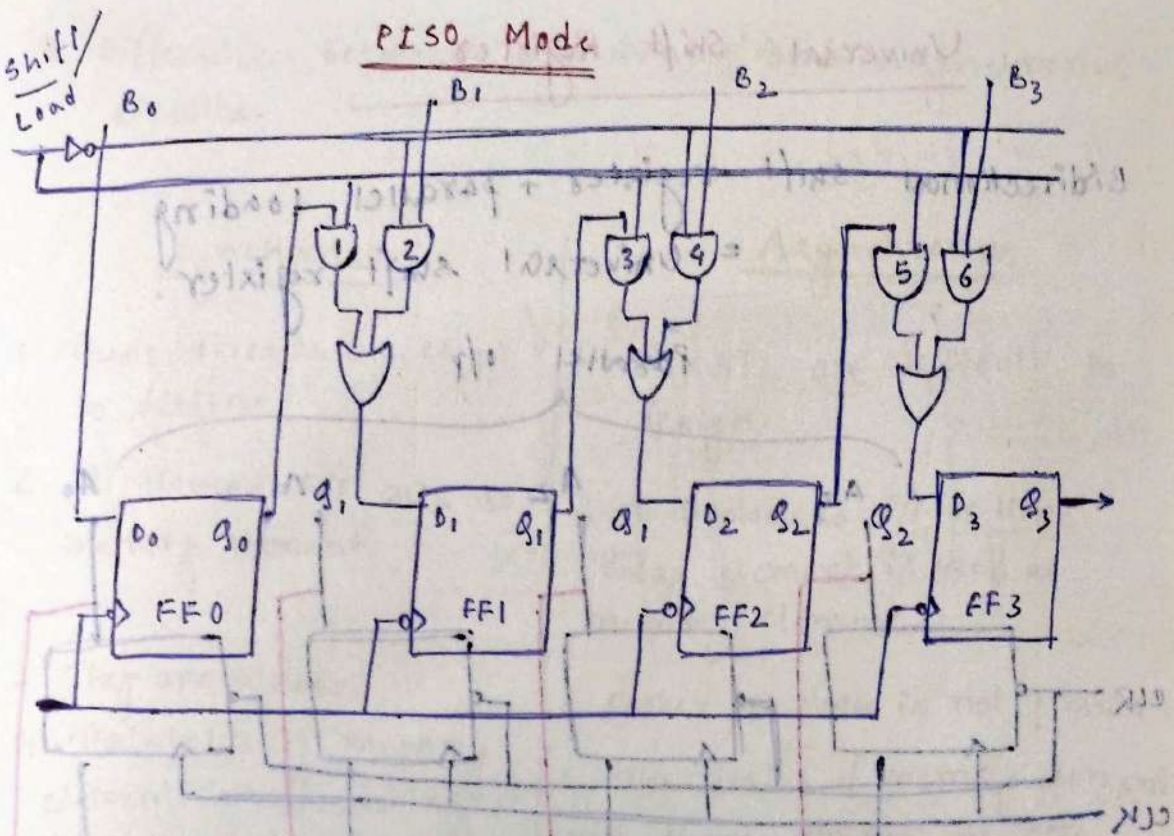
PIPO,

(Storage register / Buffer register)

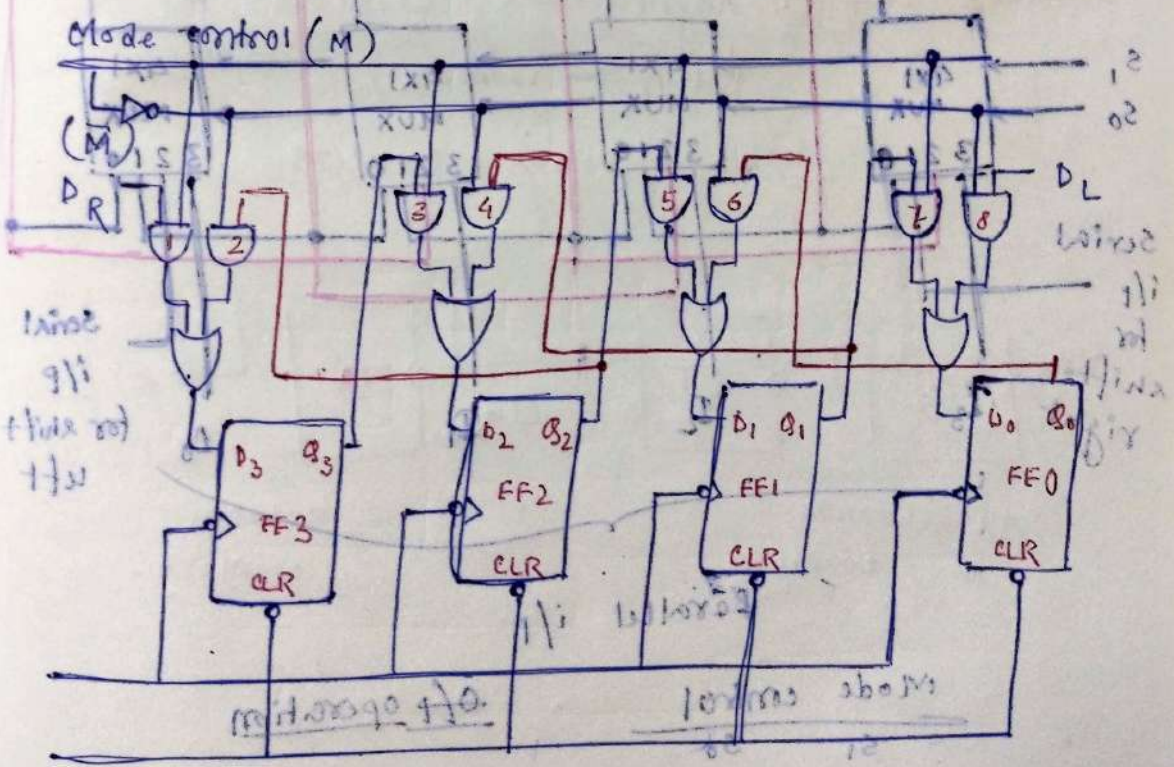



whatever we are giving input, we are getting as output. And if we remove the clock it will act as storage register.

## PISO Mode




## Bidirectional Shift Register

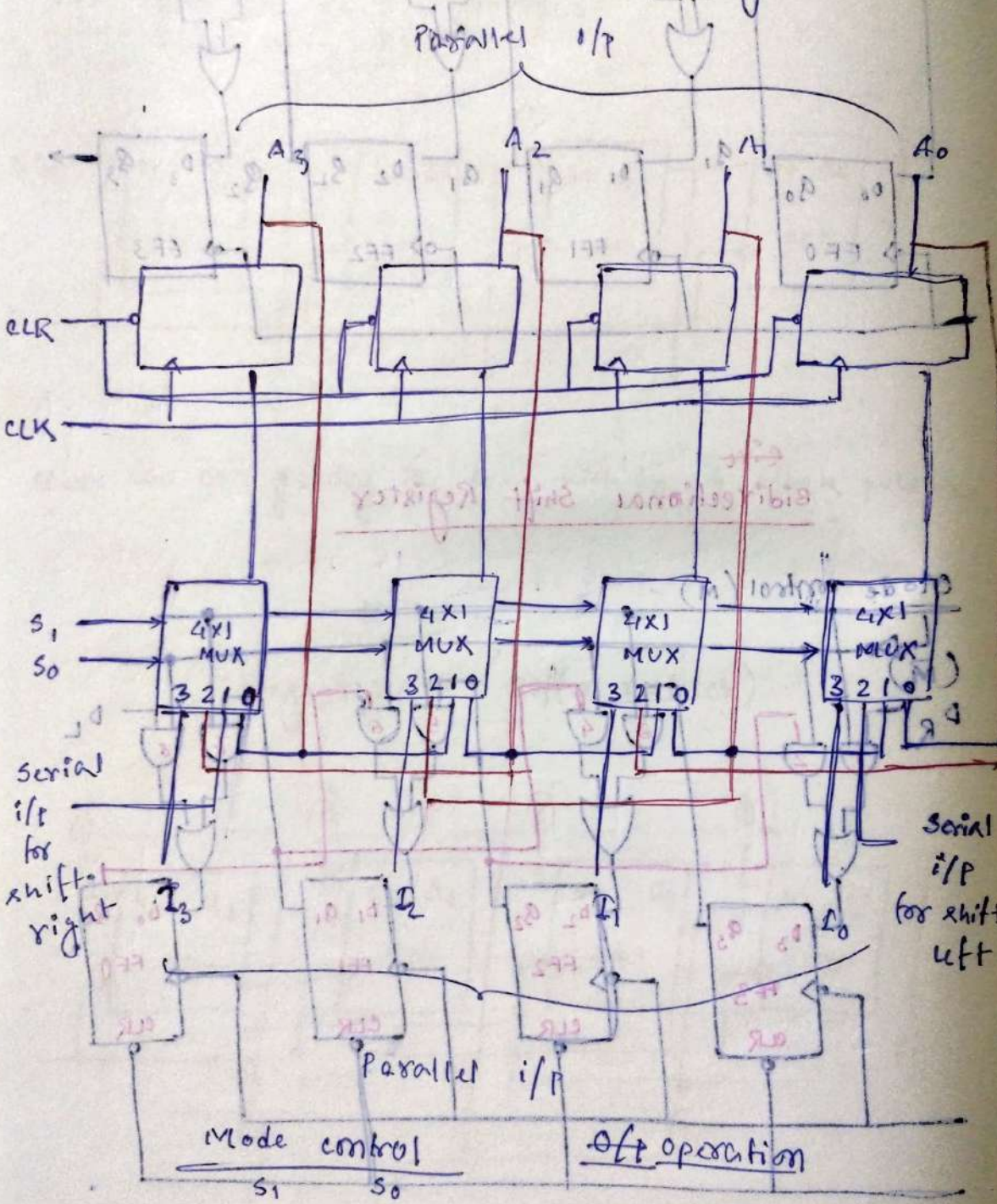



When M=1 Shift right operation

M=0 Shift left operation.

# Universal Shift Register

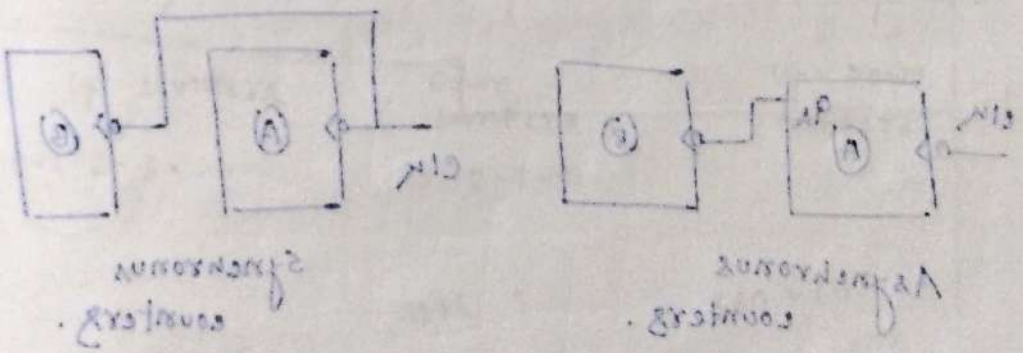
Bidirectional shift register + parallel loading
= Universal shift register.




| Mode control | | Operation |
|---|---|---|
| $S_1$ | $S_0$ | |
| 0 | 0 | No change |
| 0 | 1 | Shift right |
| 1 | 0 | Shift left |
| 1 | 1 | Parallel loading |

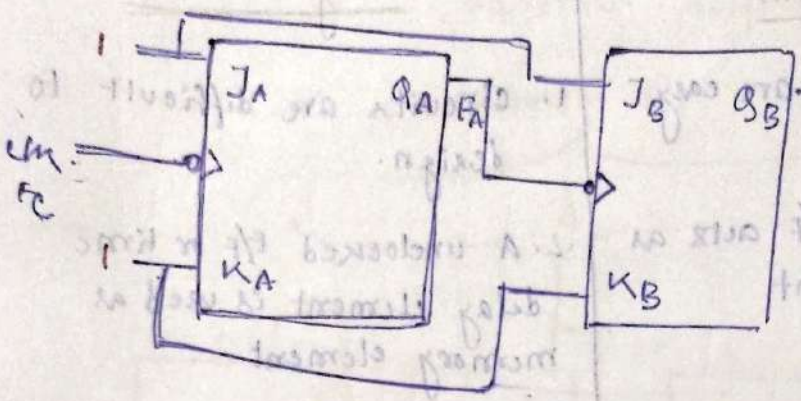
- Difference between Synchronous and Asynchronous circuits.

| Synchronous | Asynchronous |
|---|---|
| 1. These circuits are easy to design. | 1. circuits are difficult to design. |
| 2. A clocked F/F acts as memory element | 2. A unclocked F/F or time delay element is used as memory element. |
| 3. They are slower. | 3. Faster as clock is not present. |
| 4. The status of memory element is affected only at the active edge of clock, if input is changed | 4. The status of memory element will change anytime as soon as input is changed. |



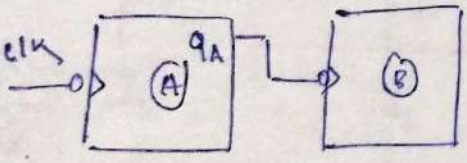
# Counters

F/F as divided by 2 circuits.

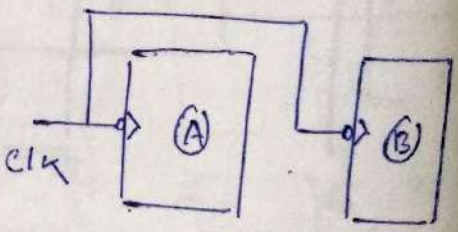



As it is a negative edge triggered FF. so all the changes will happen on its falling edge.

Counters — (i) Asynchronous Counters
(Ripple counters)
(ii) Synchronous Counters

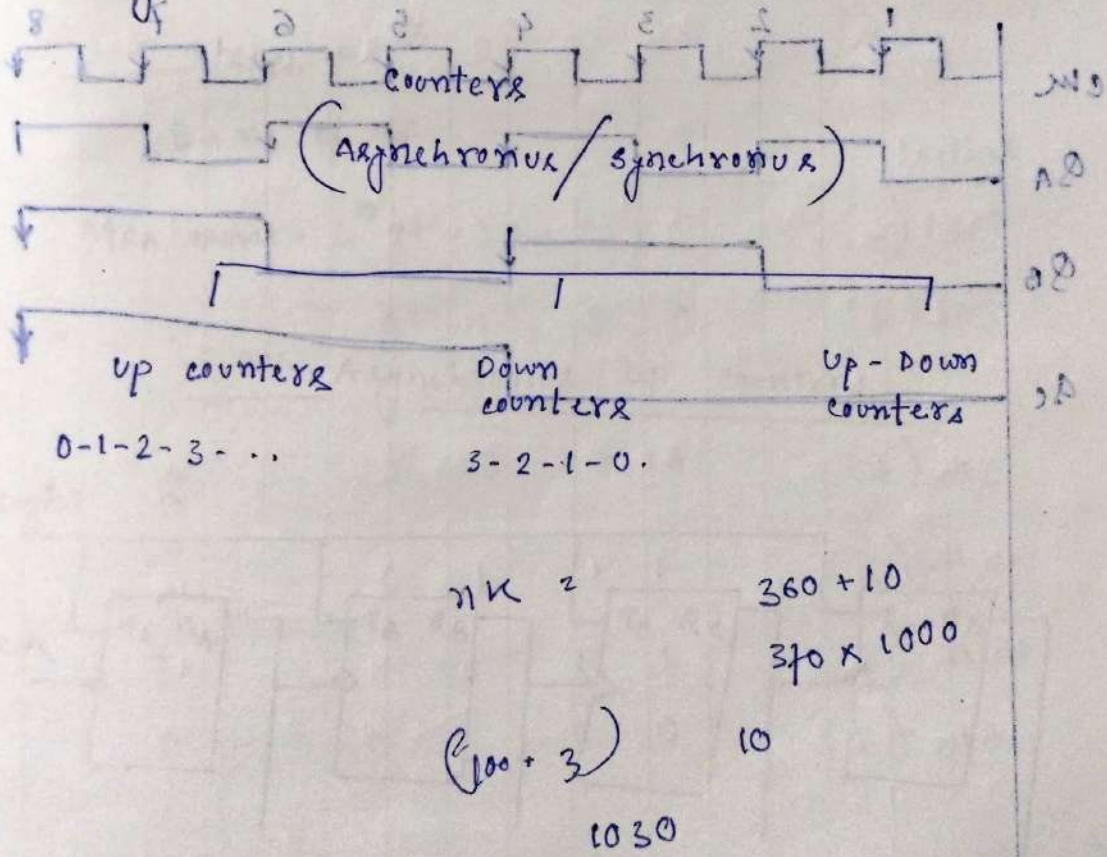



Asynchronous counters.




Synchronous counters.

| Asynchronous counter | Synchronous |
|---|---|
| 1. Flip Flops are connected in such a way that o/p of first FF drives the clock of next FF. | 1. There is no connection between o/p of first FF and clock of next flip flop. |
| 2. Flip flops are not clocked simultaneously. | 2. Flip flops are clocked simultaneously. |
| 3. circuit is simple for more no. of states. | 3. circuit becomes complicated as no. of states increases. |
| 4. Speed is slow as clock is propagated through no. of stages. | 4. Speed is high as clock is given at a same time. |



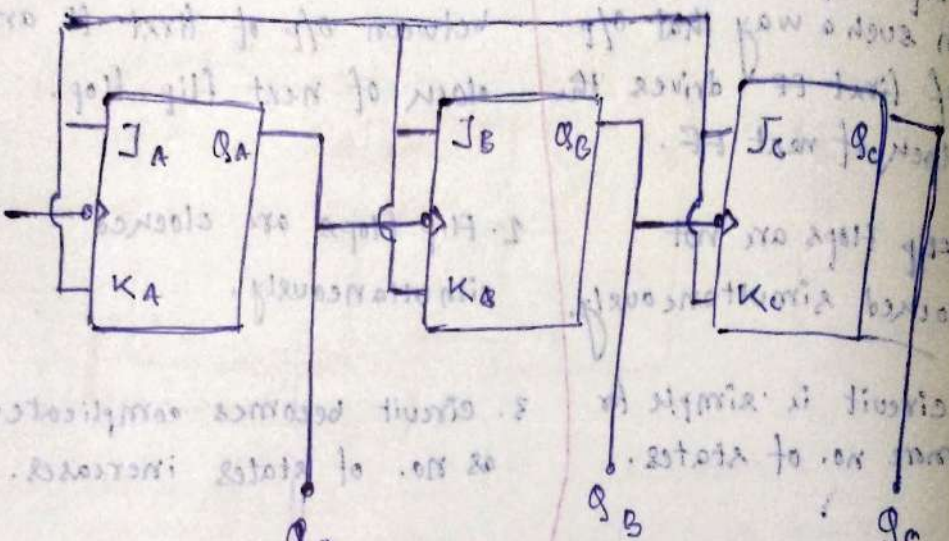

nK = 360 + 10

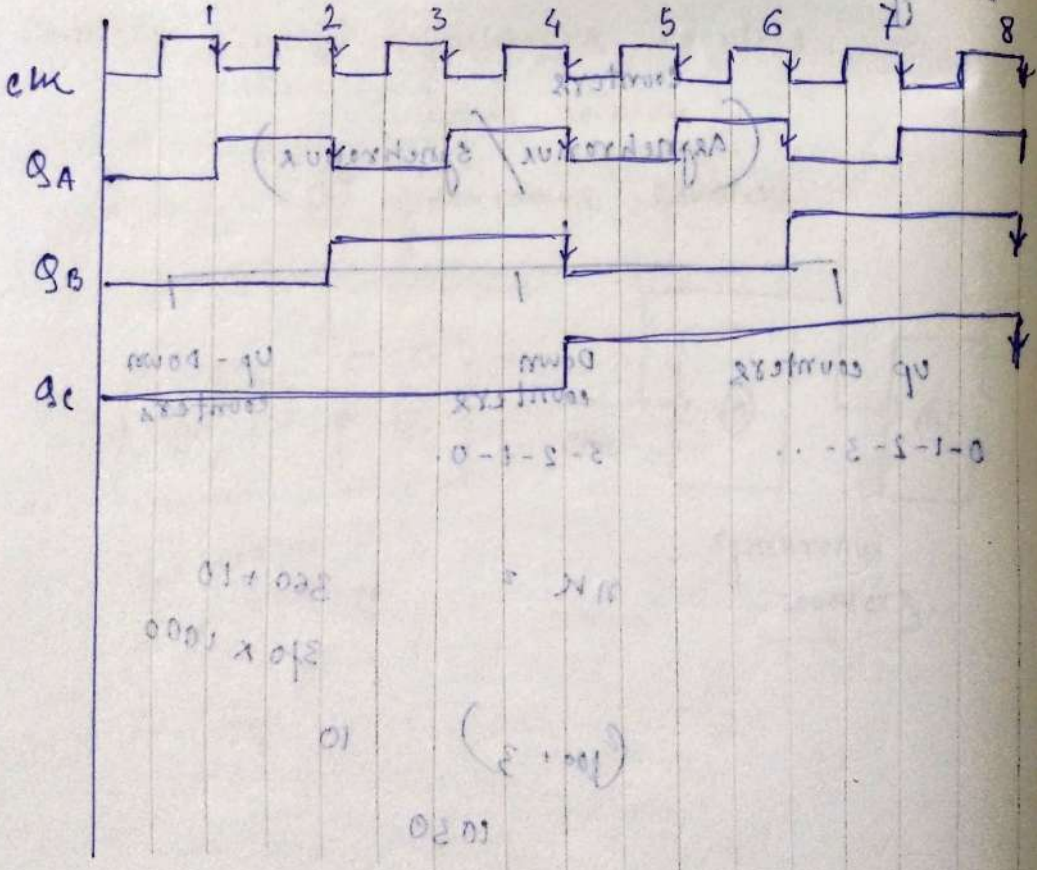
370 × 1000

(100 + 3) 10

1030

# 3 bit Asynchronous up counter



| Clock | $Q_C$ | $Q_B$ | $Q_A$ | Decimal eq. |
|---|---|---|---|---|
| Initial | 0 | 0 | 0 | 0 |
| 1st (↓) | 0 | 0 | 1 | 1 |
| 2nd (↓) | 0 | 1 | 0 | 2 |
| 3rd (↓) | 0 | 1 | 1 | 3 |
| 4th (↓) | 1 | 0 | 0 | 4 |
| 5th (↓) | 1 | 0 | 1 | 5 |
| 6th (↓) | 1 | 1 | 0 | 6 |
| 7th (↓) | 1 | 1 | 1 | 7 |
| 8th (↓) | 0 | 0 | 0 | 0 |

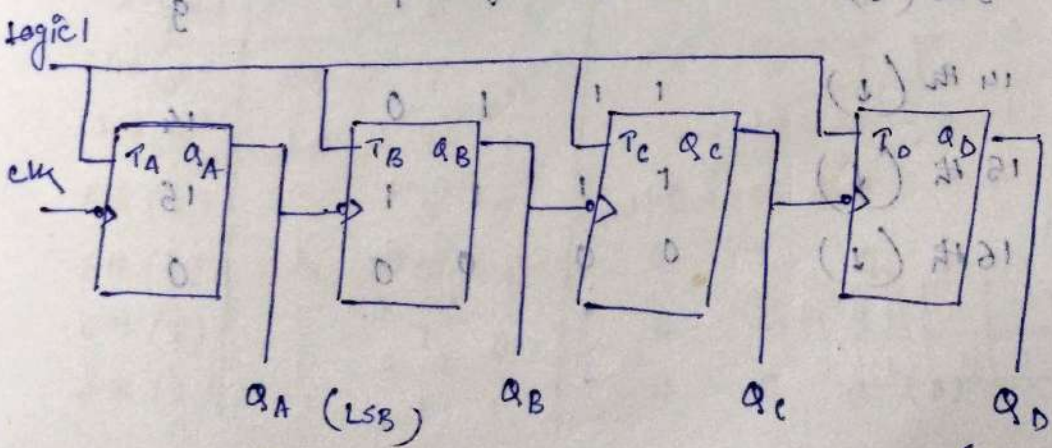
8 states

8 states $= 2^3$

3 = no. of FF.

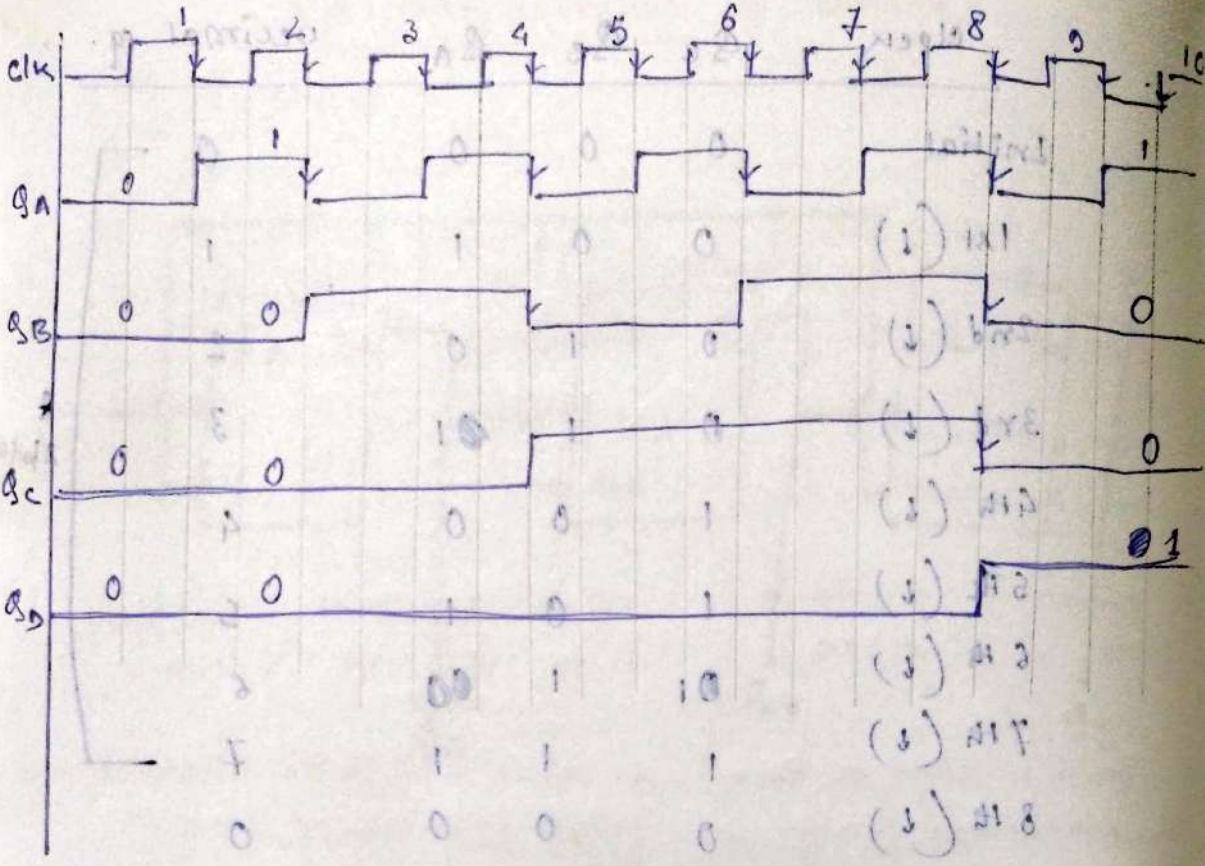
Max. count $= 2^n - 1 = 8 - 1 = 7$

## 4 Bit Asynchronous UP counters

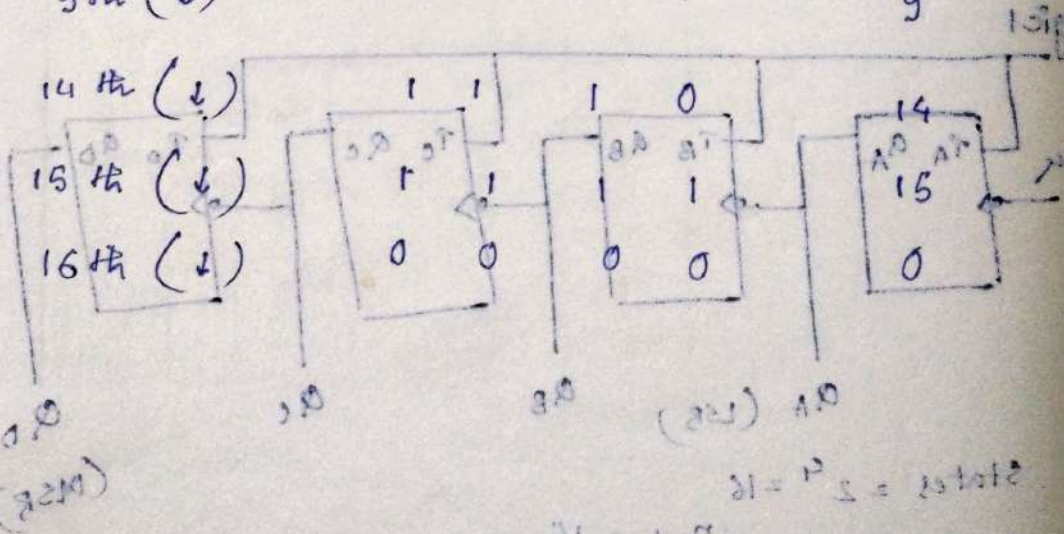



States $= 2^4 = 16$

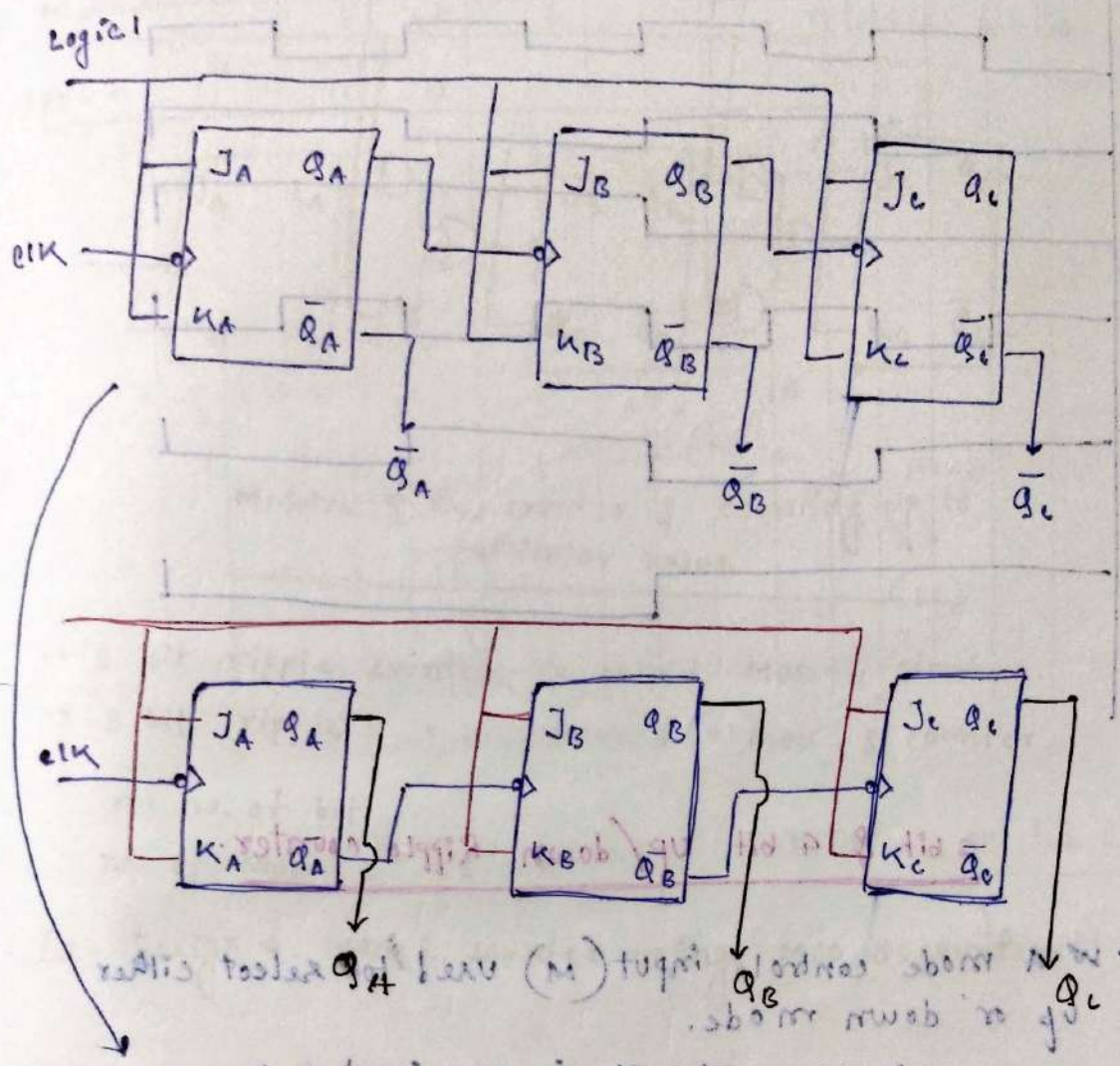
Max. count $= 2^n - 1 = 15$



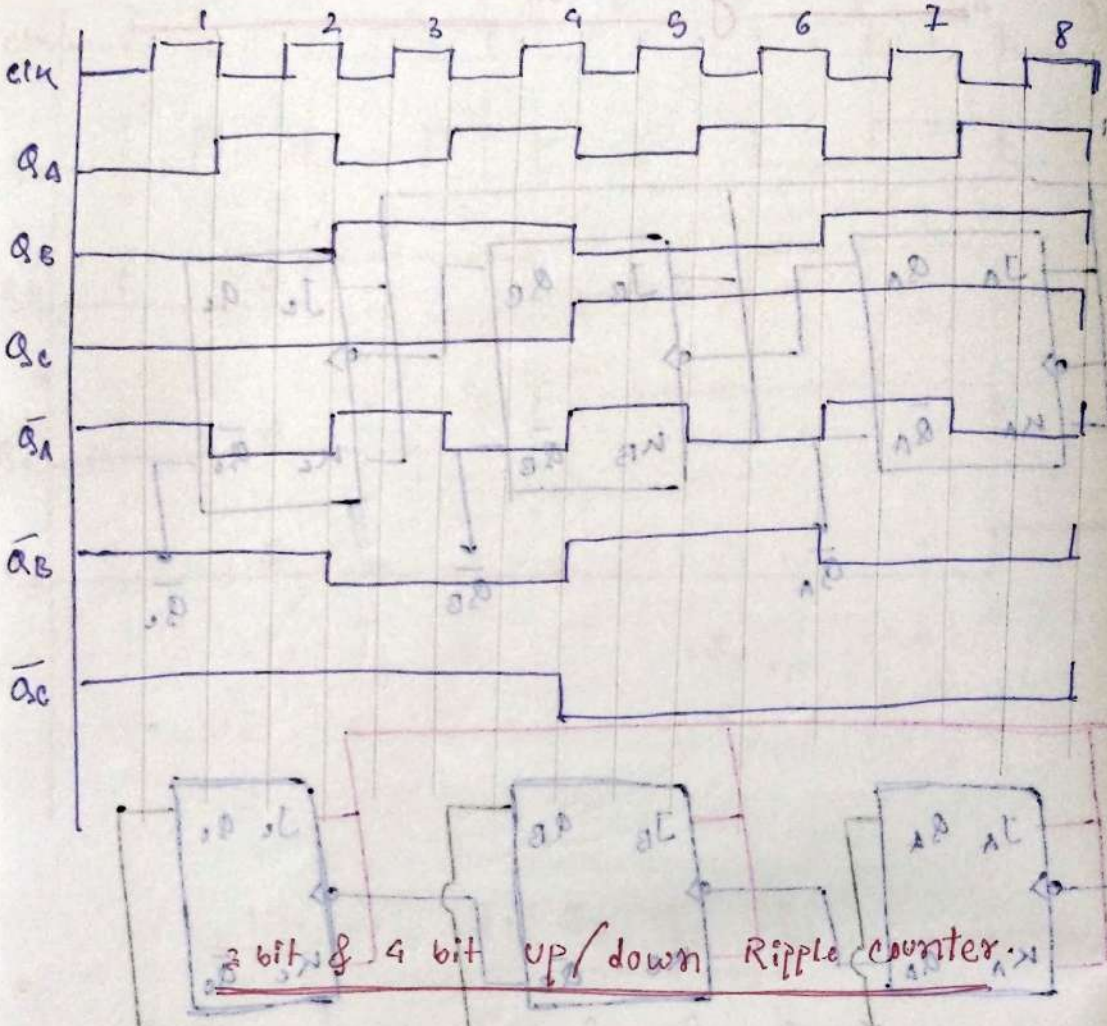

| clk | $Q_D$ | $Q_C$ | $Q_B$ | $Q_A$ | Decimal |
|---|---|---|---|---|---|
| Initial | 0 | 0 | 0 | 0 | 0 |
| 1st (↓) | 0 | 0 | 0 | 1 | 1 |
| 2nd (↓) | 0 | 0 | 1 | 0 | 2 |
| : | | | | | |
| 9th (↓) | 1 | 0 | 0 | 1 | 9 |
| 14th (↓) | 1 | 1 | 1 | 0 | 14 |
| 15th (↓) | 1 | 1 | 1 | 1 | 15 |
| 16th (↓) | 0 | 0 | 0 | 0 | 0 |



# 3 bit Asynchronous down counters




| clk | $Q_C$ | $Q_B$ | $Q_A$ | $\bar{Q}_C$ | $\bar{Q}_B$ | $\bar{Q}_A$ | |
|---|---|---|---|---|---|---|---|
| Initial | 0 | 0 | 0 | 1 | 1 | 1 | (7) |
| 1st (↓) | 0 | 0 | 1 | 1 | 1 | 0 | |
| 2nd (↓) | 0 | 1 | 0 | 1 | 0 | 1 | |
| 3rd (↓) | 0 | 1 | 1 | 1 | 0 | 0 | |
| 4th (↓) | 1 | 0 | 0 | 0 | 1 | 1 | |
| 5th (↓) | 1 | 0 | 1 | 0 | 1 | 0 | |
| 6th (↓) | 1 | 1 | 0 | 0 | 0 | 1 | |
| 7th (↓) | 1 | 1 | 1 | 0 | 0 | 0 | (0) |




3 bit & 4 bit up/down Ripple counter.

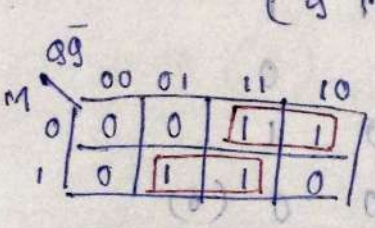
→ A mode control input (M) used to select either up or down mode.

→ A combinational circuit is required between each pair of FF.

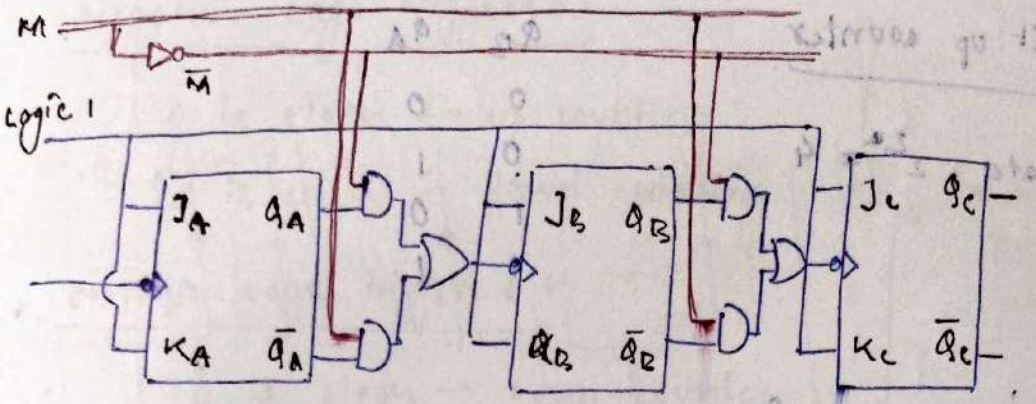
M = 0 up counting (Q connected)

M = 1 down counting. ($\bar{Q}$ connected).

| M | Q | $\bar{Q}$ | Y |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 |




$Y = \bar{M}Q + M\bar{Q}$




Modulus of the counter & counting up to particular value

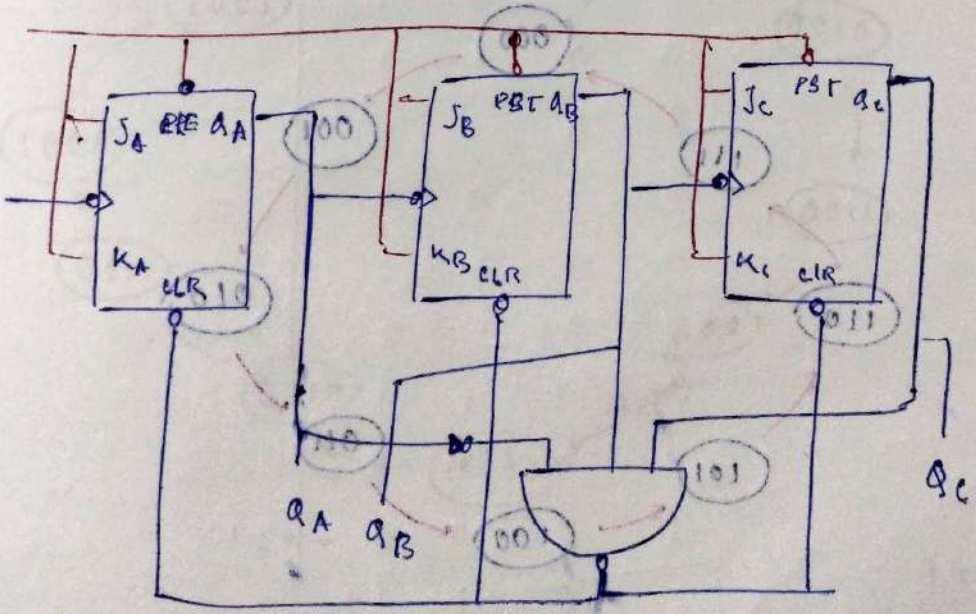
→ 2 bit ripple counter is called MOD-4 counter.

→ 3 bit ripple " " " MOD - 8 counter.

n = no. of bit

no. of MOD = $2^n = 2^2 = 4$

Ex. Design a MOD 6 counter using MOD 8 counters.




PST = 0 Q = 1

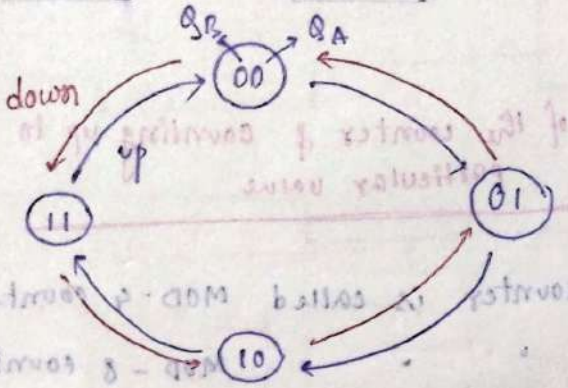
CLR = 0 Q = 0

# State Diagram of counter

## 2 bit up counter

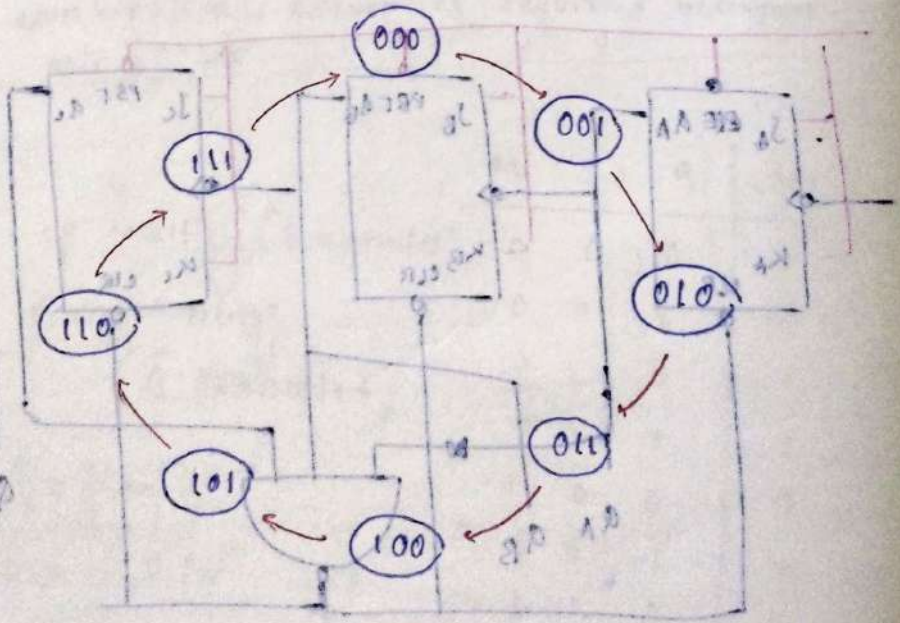
| $Q_B$ | $Q_A$ |
|---|---|
| 0 | 0 |
| 0 | 1 |
| 1 | 0 |
| 1 | 1 |

States $= 2^{2_{no.}} = 4$




## 3 bit up counter :

No. of states $= 2^3 = 8$



# Decade (BCD) Ripple counter

**Negative edge triggered:**

Q is clock → up counting

$\bar{Q}$ is clock → down counting.

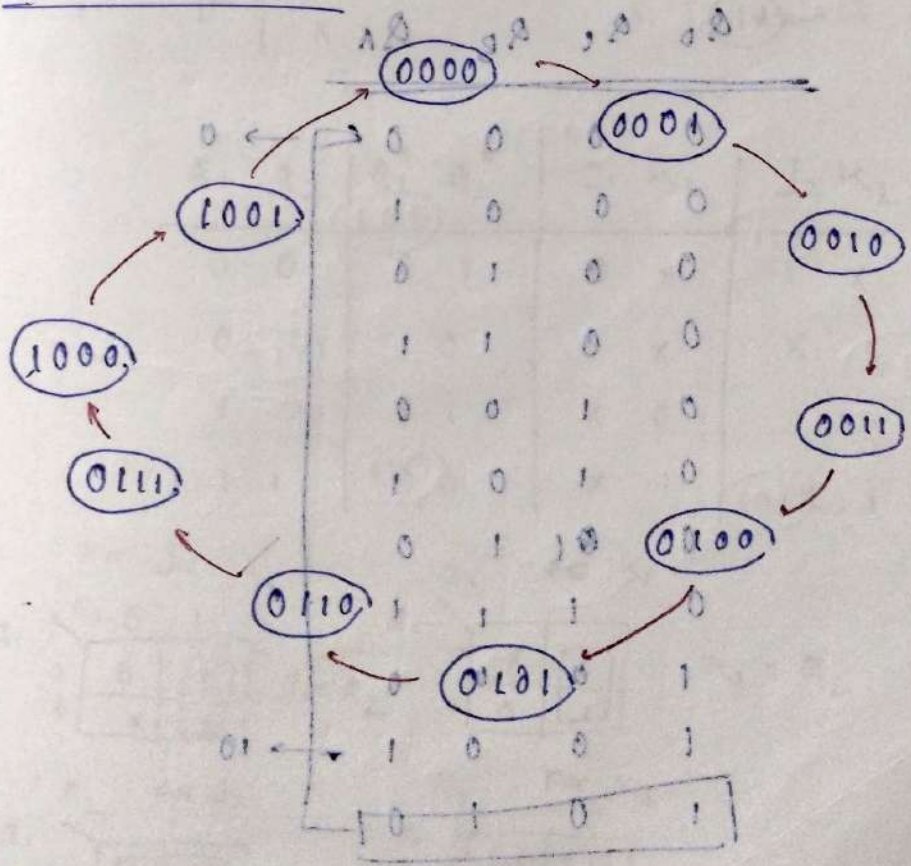
**Positive edge triggered:**

Q is clock → down counter

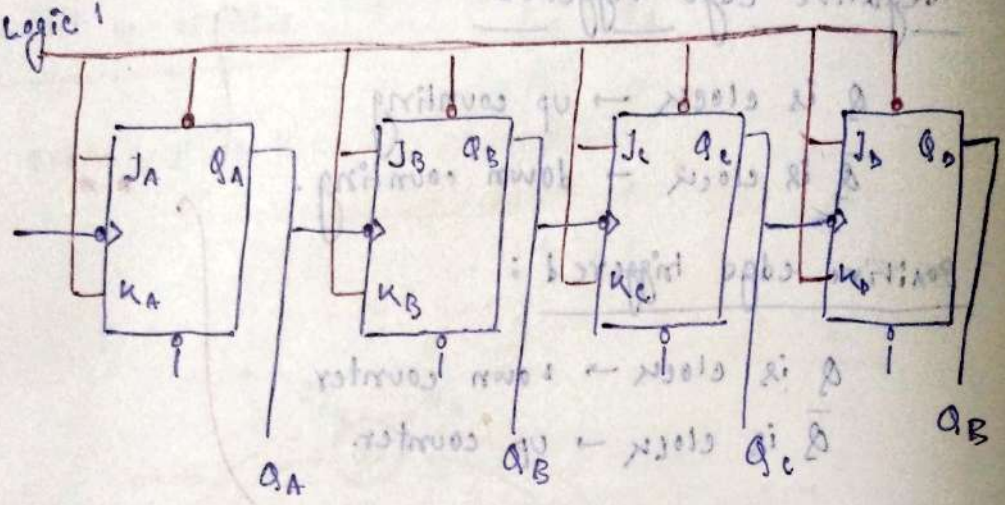
$\bar{Q}$ is clock → UP counter.

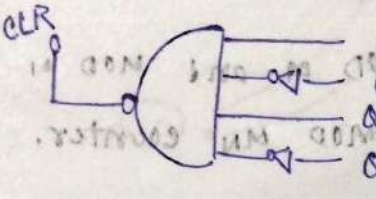
If we cascade MOD M and MOD N counter we will get MOD MN counter.

No. of state = 10



so we have to design MOD-10 using MOD-16




| $Q_D$ | $Q_C$ | $Q_B$ | $Q_A$ | |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | → 0 |
| 0 | 0 | 0 | 1 | |
| 0 | 0 | 1 | 0 | |
| 0 | 0 | 1 | 1 | |
| 0 | 1 | 0 | 0 | |
| 0 | 1 | 0 | 1 | |
| 0 | 1 | 1 | 0 | |
| 0 | 1 | 1 | 1 | |
| 1 | 0 | 0 | 0 | |
| 1 | 0 | 0 | 1 | → 10 |
| 1 | 0 | 1 | 0 | |

# How to Design Synchronous Counters

Step 1 : Decide the no. of FF

Step 2 : Excitation table of FF

Step 3 : State diagram and circuit excitation table.

Step 4 : Obtain simplified eqn using K-map.

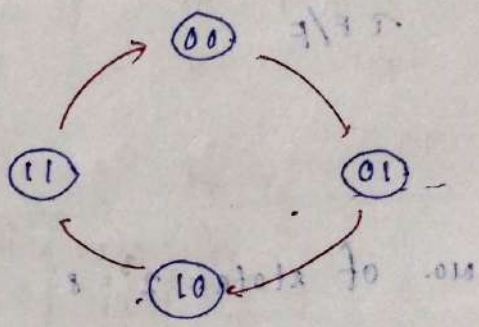
Step 5 : Draw logic diagram.

## # Design 2 bit synchronous up counter.

J-K FF

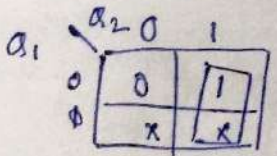
| $Q_n$ | $Q_{n+1}$ | J | K |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |

States = $2^2 = 4$



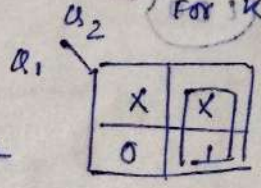

| $Q_1$ | $Q_2$ | $Q_1^*$ | $Q_2^*$ | $J_1$ | $K_1$ | $J_2$ | $K_2$ |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | X | 1 | X |
| 0 | 1 | 1 | 0 | 1 | X | X | 1 |
| 1 | 0 | 1 | 1 | X | 0 | 1 | X |
| 1 | 1 | 0 | 0 | X | 1 | X | 1 |

For $J_1$



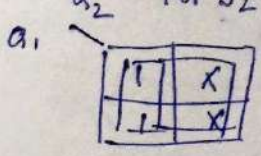

$J_1 = Q_2$

For $K_1$



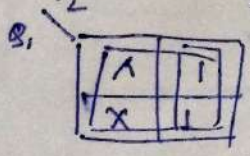

$K_1 = Q_2$

For $J_2$

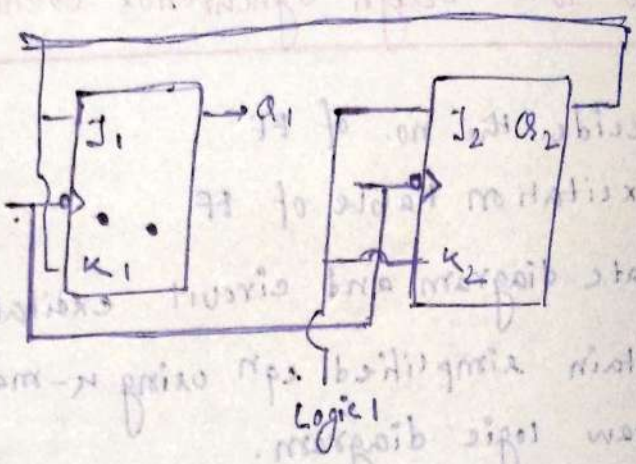



$J_2 = \overline{Q}_1$

For $K_2$




$K_2 = Q_1$



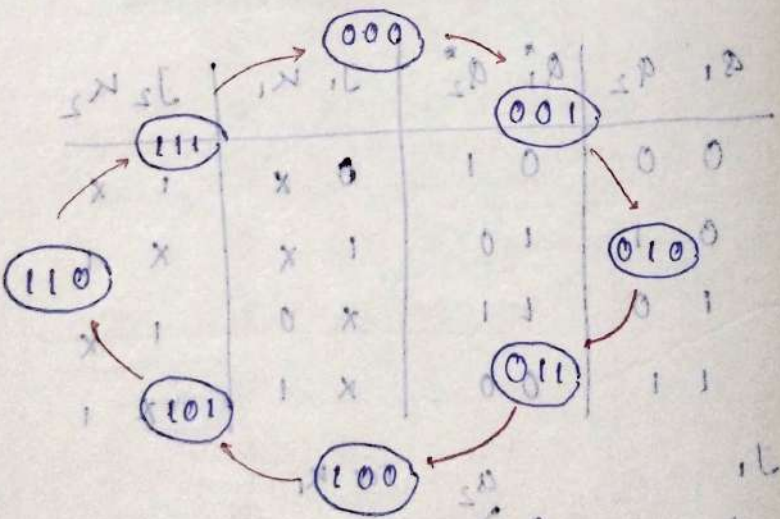

## 3 bit synchronous up counter

No. of F/F = 3

∴ 3 F/F

No. of state = $2^3 = 8$

| $Q_n$ | $Q_{n+1}$ | T |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |



| P.S | | | N.S | | | | | |
|---|---|---|---|---|---|---|---|---|
| $Q_C$ | $Q_B$ | $Q_A$ | $Q_C^+$ | $Q_B^+$ | $Q_A^+$ | $T_C$ | $T_B$ | $T_A$ |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |

For $T_C$

| $Q_C$ \ $Q_BQ_A$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 |

$T_C = Q_BQ_A$

For $T_B$

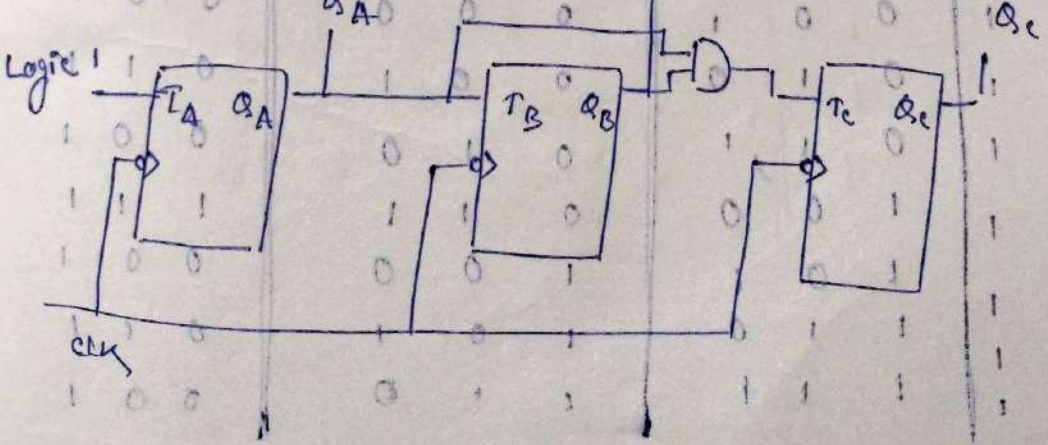
| $Q_C$ \ $Q_BQ_A$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 1 | 1 | 0 |

$T_B = Q_A$

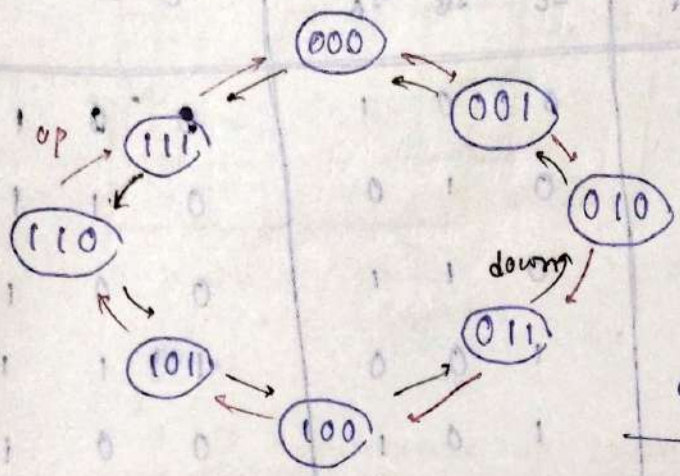
For $T_A$

| $Q_C$ \ $Q_BQ_A$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 |

$T_A = 1$



# 3 bit up/down synchronous counter




| $Q_n$ | $Q_{n+1}$ | T |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

M = 0 up counter

M = 1 down counter.

| M | $Q_C$ | $Q_B$ | $Q_A$ | $Q_C^*$ | $Q_B^*$ | $Q_A^*$ | $T_C$ | $T_B$ | $T_A$ |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 |

For $T_C$

| $MQ_C$ \ $Q_BQ_A$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 0 | 1 | 0 |
| 01 | 0 | 0 | 1 | 0 |
| 11 | 1 | 0 | 0 | 0 |
| 10 | 1 | 0 | 0 | 0 |

$T_C = \bar{M} Q_B Q_A + M \bar{Q}_B \bar{Q}_A$

For $T_B$

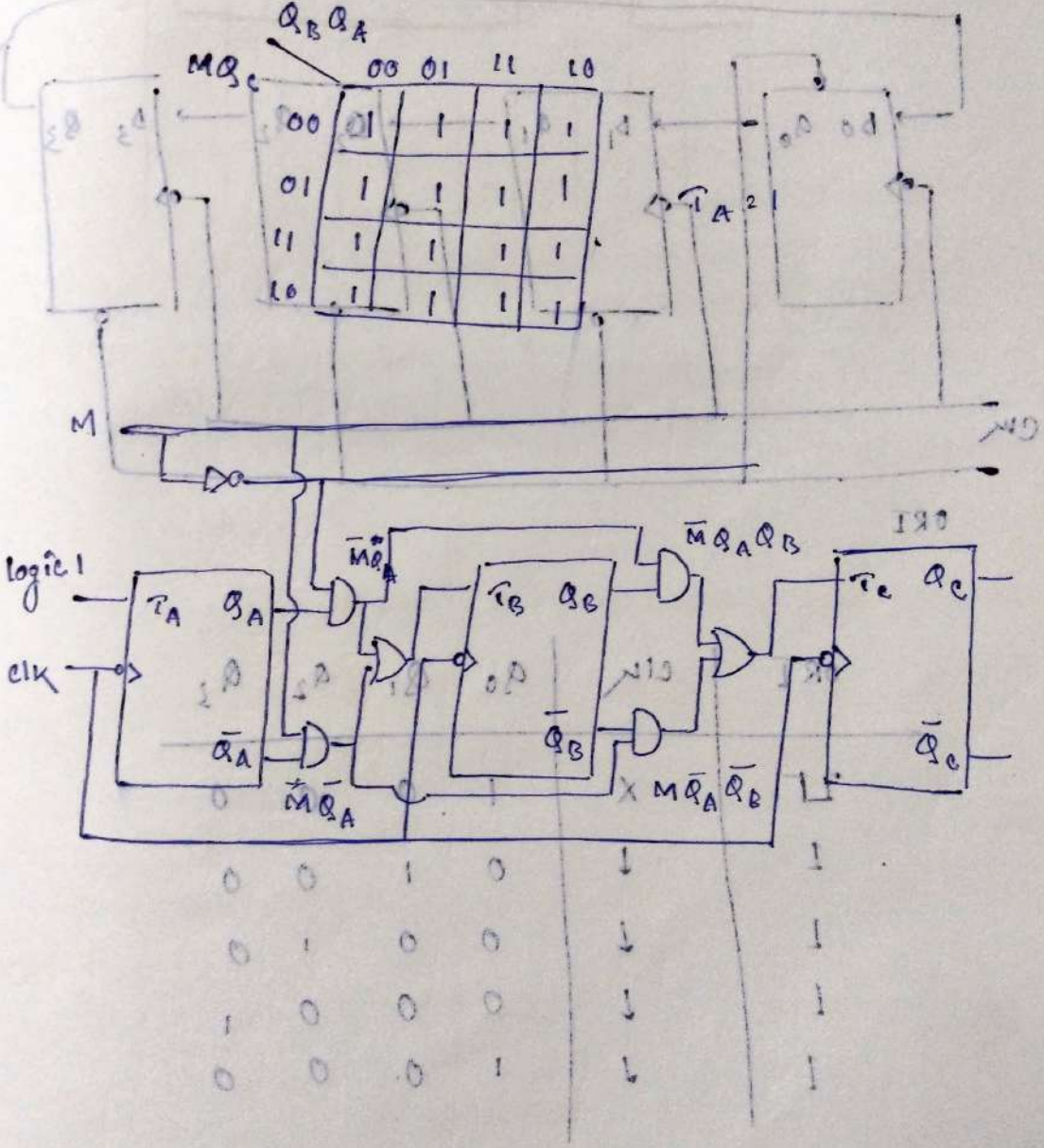
| $MQ_C$ \ $Q_BQ_A$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 0 | 1 | 1 | 0 |
| 01 | 0 | 1 | 1 | 0 |
| 11 | 1 | 0 | 0 | 1 |
| 10 | 1 | 0 | 0 | 1 |

$T_B = \bar{M} Q_A + M Q_A$

$= M \oplus Q_A$

| $MQ_C$ \ $Q_BQ_A$ | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 00 | 1 | 1 | 1 | 1 |
| 01 | 1 | 1 | 1 | 1 |
| 11 | 1 | 1 | 1 | 1 |
| 10 | 1 | 1 | 1 | 1 |

$T_A = 1$



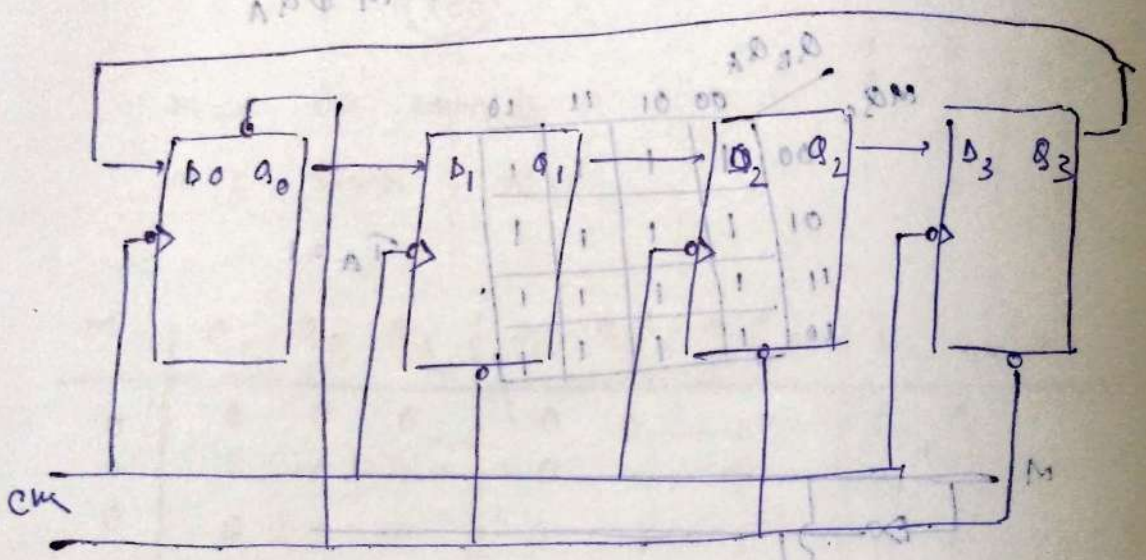
# Ring Counters

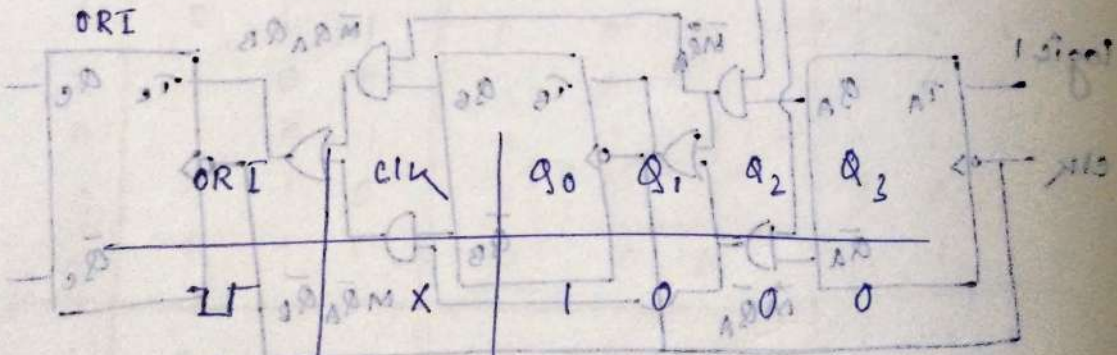
→ Ring counter is a typical application of shift register.

→ The only change is the o/p of last FF is connected to the input of first F/F.

→ No. of states = no. of F/F used.

**




| ORI | CLK | $Q_0$ | $Q_1$ | $Q_2$ | $Q_3$ |
|---|---|---|---|---|---|
| 1 | X | 1 | 0 | 0 | 0 |
| 1 | ↓ | 0 | 1 | 0 | 0 |
| 1 | ↓ | 0 | 0 | 1 | 0 |
| 1 | ↓ | 0 | 0 | 0 | 1 |
| 1 | ↓ | 1 | 0 | 0 | 0 |